Topology II

Lecture notes, Chapters 9–14 (Algebraic Topology)

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Chapter 9. The Fundamental Group

9.1 Homotopy of paths

Motivation. Point-set topology cannot tell the plane from the punctured plane by counting components or checking compactness. What distinguishes them is a loop around the hole that cannot be shrunk to a point. Homotopy makes "deforming one path into another" precise, and it is the starting point of algebraic topology: turning shapes into groups.

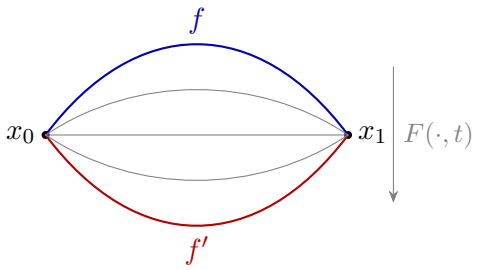


Figure. *A path homotopy F deforms f into f' through paths $F(\cdot,t)$ that all keep the end points x_0, x_1 fixed.*

Maps $f, f' : X \to Y$ are **homotopic**, $f \simeq f'$, if there is a continuous $F : X \times I \to Y$ ($I = [0,1]$) with $F(x,0) = f(x)$ and $F(x,1) = f'(x)$; f is **nulhomotopic** if it is homotopic to a constant map. Paths $f, f' : I \to X$ from x_0 to x_1 are **path homotopic**, $f \simeq_p f'$, if there is $F : I \times I \to X$ with $F(s,0) = f(s)$, $F(s,1) = f'(s)$, $F(0,t) = x_0$, $F(1,t) = x_1$.

Lemma 9.1 (51.1)

$\simeq$ and $\simeq_p$ are equivalence relations.

Proof. $F(x,t) = f(x)$ gives $f \simeq f$; if F is a homotopy from f to f', then $F(x,1-t)$ is one from f' to f. If $F : f \simeq f'$ and $F' : f' \simeq f''$, define $G(x,t) = F(x,2t)$ for $t \le \frac{1}{2}$ and $F'(x,2t-1)$ for $t \ge \frac{1}{2}$; the two formulas agree at $t = \frac{1}{2}$, so G is continuous by the pasting lemma. For paths, all these homotopies keep the end points fixed. ∎

We write $[f]$ for the path-homotopy class of f.

Example 9.1

If $f, g : X \to \mathbb{R}^2$ are maps (or paths with the same end points), the **straight-line homotopy** $F(x,t) = (1-t)f(x) + tg(x)$ is a homotopy (path homotopy). More generally this works for maps into any convex subset of $\mathbb{R}^n$. In $\mathbb{R}^2 - \{0\}$, however, the paths $f(s) = (\cos \pi s, \sin \pi s)$ and $g(s) = (\cos \pi s, -\sin \pi s)$ are not path homotopic (Section 9.4).

If f is a path from x_0 to x_1 and g a path from x_1 to x_2, the **product** $f * g$ is $f(2s)$ on $[0, \frac{1}{2}]$ and $g(2s-1)$ on $[\frac{1}{2}, 1]$. It is well defined on classes: if $F : f \simeq_p f'$ and $G : g \simeq_p g'$, then $H(s,t) = F(2s,t)$ for $s \le \frac{1}{2}$, $G(2s-1,t)$ for $s \ge \frac{1}{2}$, is a path homotopy $f * g \simeq_p f' * g'$. So $[f] * [g] = [f * g]$ is defined. Two observations will be used repeatedly: if $k : X \to Y$ is continuous and F is a path homotopy from f to f', then $k \circ F$ is a path homotopy from $k \circ f$ to $k \circ f'$; and $k \circ (f * g) = (k \circ f) * (k \circ g)$.

Theorem 9.2 (51.2)

The operation $*$ on path-homotopy classes satisfies:
(1) (associativity) if $[f]*([g]*[h])$ is defined, so is $([f]*[g])*[h]$, and they are equal;
(2) (identities) if e_x is the constant path at x and f runs from x_0 to x_1, then $[e_{x_0}]*[f] = [f] = [f]*[e_{x_1}]$;
(3) (inverses) with $\bar{f}(s) = f(1-s)$, $[f]*[\bar{f}] = [e_{x_0}]$ and $[\bar{f}]*[f] = [e_{x_1}]$.

Proof. Let e_0 be the constant path in I at 0, $i : I \to I$ the identity path, and $\bar{i}$ its reverse.

(2) $e_0 * i$ and i are paths in the convex set I from 0 to 1, so they are path homotopic by a straight-line homotopy G; then $f \circ G$ is a path homotopy between $f \circ (e_0 * i) = e_{x_0} * f$ and $f \circ i = f$. The other identity uses $i * e_1$.

(3) $i * \bar{i}$ and e_0 are paths in I from 0 to 0, hence path homotopic by some H; then $f \circ H$ shows $f * \bar{f} \simeq_p e_{x_0}$. Similarly $\bar{f} * f \simeq_p e_{x_1}$.

(1) For $0 < a < b < 1$ let $k_{a,b}$ be the path that on $[0,a]$, $[a,b]$, $[b,1]$ is the positive linear map of that interval onto I followed by f, g, h respectively. Then $(f*g)*h = k_{1/2,3/4}$ and $f*(g*h) = k_{1/4,1/2}$. For any a, b, c, d, let $\gamma : I \to I$ be the piecewise linear map sending $0, a, b, 1$ to $0, c, d, 1$; then $k_{c,d} \circ \gamma = k_{a,b}$. Since γ and i are paths in I from 0 to 1, $\gamma \simeq_p i$, and composing with $k_{c,d}$ gives $k_{a,b} \simeq_p k_{c,d}$. ■

Theorem 9.3 (51.3)

Let f be a path in X and $0 = a_0 < a_1 < \cdots < a_n = 1$. Let f_i be the positive linear map of I onto $[a_{i-1}, a_i]$ followed by f. Then $[f] = [f_1] * \cdots * [f_n]$.

Proof. By associativity the right side is well defined. Represent it by a specific bracketing of $f_1 * \cdots * f_n$; this path is defined on a subdivision $0 = b_0 < \cdots < b_n = 1$ and on $[b_{i-1}, b_i]$ is the linear map onto I followed by f_i. Hence it equals $f \circ \gamma$, where $\gamma : I \to I$ is the piecewise linear map sending each $[b_{i-1}, b_i]$ linearly onto $[a_{i-1}, a_i]$. Since $\gamma \simeq_p i$ in I, $f \circ \gamma \simeq_p f$. ■

9.2 The fundamental group

Motivation. Loops at a base point, up to homotopy, form a group under concatenation. Homeomorphic spaces have isomorphic fundamental groups, so the group is an invariant that can prove two spaces are different.

For $x_0 \in X$, a **loop** at x_0 is a path from x_0 to x_0. By Theorem 9.2 the path-homotopy classes of loops at x_0 form a group under $*$, the **fundamental group** $\pi_1(X, x_0)$. If X is a convex subset of $\mathbb{R}^n$, $\pi_1(X, x_0)$ is trivial (straight-line homotopy to the constant loop). For a path α from x_0 to x_1 define $\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$ by $\hat{\alpha}([f]) = [\bar{\alpha}] * [f] * [\alpha]$.

Theorem 9.4 (52.1)

$\hat{\alpha}$ is a group isomorphism.

Proof. $\hat{\alpha}([f]) * \hat{\alpha}([g]) = [\bar{\alpha}] * [f] * [\alpha] * [\bar{\alpha}] * [g] * [\alpha] = \hat{\alpha}([f] * [g])$. With $\beta = \bar{\alpha}$, $\hat{\beta}(\hat{\alpha}([h])) = [\alpha] * [\bar{\alpha}] * [h] * [\alpha] * [\bar{\alpha}] = [h]$, and similarly $\hat{\alpha} \circ \hat{\beta} = \mathrm{id}$. ■

Corollary 9.5 (52.2)

If X is path connected and $x_0, x_1 \in X$, then $\pi_1(X, x_0) \cong \pi_1(X, x_1)$.

If C is the path component of x_0, then $\pi_1(C, x_0) = \pi_1(X, x_0)$, since loops and homotopies based at x_0 lie in C. A path-connected space with trivial fundamental group is **simply connected**.

Lemma 9.6 (52.3)

In a simply connected space, any two paths with the same initial and final points are path homotopic.

Proof. If f, g run from x_0 to x_1, then $f * \bar{g}$ is a loop at x_0, so $[f] * [\bar{g}] = [e_{x_0}]$; multiplying on the right by $[g]$ gives $[f] = [g]$. ■

A continuous $h : (X, x_0) \to (Y, y_0)$ (that is, $h(x_0) = y_0$) induces $h_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$, $h_*([f]) = [h \circ f]$, well defined and a homomorphism by the observations before Theorem 9.2.

Theorem 9.7 (52.4)

If $h : (X, x_0) \to (Y, y_0)$ and $k : (Y, y_0) \to (Z, z_0)$ are continuous, then $(k \circ h)_* = k_* \circ h_*$. If i is the identity of (X, x_0), then i_* is the identity.

Proof. $(k \circ h)_*([f]) = [(k \circ h) \circ f] = [k \circ (h \circ f)] = k_*(h_*([f]))$, and $i_*([f]) = [f]$. ■

Corollary 9.8 (52.5)

If $h : (X, x_0) \to (Y, y_0)$ is a homeomorphism, h_* is an isomorphism.

Proof. With $k = h^{-1}$, $k_* h_* = (k \circ h)_* = \mathrm{id}$ and $h_* k_* = \mathrm{id}$. ■

9.3 Covering spaces

Motivation. To compute a fundamental group we need a way to tell loops apart. A covering space unwraps the space so that loops lift to paths, and the endpoint of the lifted path detects how the loop winds; $\mathbb{R} \to S^1$, $t \mapsto e^{2\pi i t}$ is the model example.

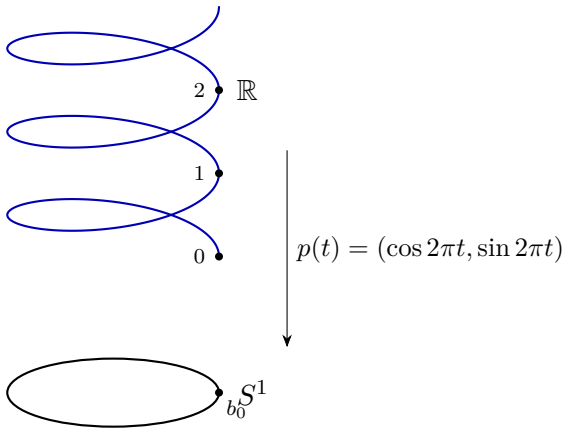


Figure. *The covering map $p : \mathbb{R} \to S^1$ wraps the line around the circle; the fibre $p^{-1}(b_0) = \mathbb{Z}$. A loop at b_0 lifts to a path from 0 to an integer, and that integer is the loop's class in $\pi_1(S^1) \cong \mathbb{Z}$.*

Let $p : E \to B$ be continuous and surjective. An open $U \subset B$ is **evenly covered** by p if $p^{-1}(U)$ is a union of disjoint open sets V_α (the **slices**), each mapped homeomorphically onto U by p. If every point of B has an evenly covered neighbourhood, p is a **covering map** and E a **covering space** of B. A covering map is a local homeomorphism and an open map, and each fibre $p^{-1}(b)$ is discrete (each of its points is the only one in its slice).

Theorem 9.9 (53.1)

$p : \mathbb{R} \to S^1$, $p(x) = (\cos 2\pi x, \sin 2\pi x)$, is a covering map.

Proof. Let U be the open arc of points with positive first coordinate. Then $p^{-1}(U) = \bigcup_n (n - \frac{1}{4}, n + \frac{1}{4})$. On $[n - \frac{1}{4}, n + \frac{1}{4}]$, p is injective (its second coordinate is strictly increasing), so it maps this compact interval homeomorphically onto the closed right half circle; restricting, $(n - \frac{1}{4}, n + \frac{1}{4})$ maps homeomorphically onto U. The same argument applies to the arcs with negative first coordinate, positive second coordinate and negative second coordinate, and these four arcs cover S^1. ■

Example 9.2

The restriction of p to $(0, \infty)$ is a surjective local homeomorphism but not a covering map: no neighbourhood of $b_0 = (1, 0)$ is evenly covered, because the component of $p^{-1}(U)$ containing a point near 0 is a half-open interval $(0, \epsilon)$ that is not mapped onto U.

Theorem 9.10 (53.2)

If $p : E \to B$ is a covering map, $B_0 \subset B$ and $E_0 = p^{-1}(B_0)$, then $p_0 = p|E_0 : E_0 \to B_0$ is a covering map.

Proof. For $b \in B_0$ let U be evenly covered with slices V_α. Then $U \cap B_0$ is a neighbourhood of b in B_0, and $p_0^{-1}(U \cap B_0)$ is the disjoint union of the sets $V_\alpha \cap E_0$, open in E_0, each mapped homeomorphically onto $U \cap B_0$. ■

Theorem 9.11 (53.3)

If $p : E \to B$ and $p' : E' \to B'$ are covering maps, so is $p \times p' : E \times E' \to B \times B'$.

Proof. If U, U' are evenly covered neighbourhoods of b, b' with slices V_α, V'_β, then $(p \times p')^{-1}(U \times U')$ is the disjoint union of the open sets $V_\alpha \times V'_\beta$, each mapped homeomorphically onto $U \times U'$. ■

Example 9.3

$p \times p : \mathbb{R} \times \mathbb{R} \to S^1 \times S^1$ is a covering of the torus $T = S^1 \times S^1$. Restricting to $b_0 \times S^1 \cup S^1 \times b_0$ (Theorem 9.10) gives a covering of the figure eight $b_0 \times S^1 \cup S^1 \times b_0$ by the grid of horizontal and vertical lines through integer points. Since $\mathbb{R}_+ \times S^1 \cong \mathbb{R}^2 - 0$ by polar coordinates, $\mathrm{id} \times p : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}^2 - 0$ is a covering of the punctured plane.

9.4 The fundamental group of the circle

Motivation. Using the covering $\mathbb{R} \to S^1$ we prove $\pi_1(S^1) \cong \mathbb{Z}$, the first nontrivial computation. The rest of the chapter is a series of applications of this one result.

If $p : E \to B$ is a map, a **lifting** of $f : X \to B$ is a map $\tilde{f} : X \to E$ with $p \circ \tilde{f} = f$.

Lemma 9.12 (54.1)

Let $p : E \to B$ be a covering map with $p(e_0) = b_0$. Any path $f : I \to B$ beginning at b_0 has a unique lifting $\tilde{f}$ beginning at e_0.

Proof. Cover B by evenly covered open sets; by the Lebesgue number lemma there is a subdivision $0 = s_0 < \cdots < s_n = 1$ such that each $f([s_i, s_{i+1}])$ lies in an evenly covered U.

Existence: set $\tilde{f}(0) = e_0$; if $\tilde{f}$ is defined on $[0, s_i]$, let V_0 be the slice of $p^{-1}(U)$ containing $\tilde{f}(s_i)$ and define $\tilde{f}(s) = (p|V_0)^{-1}(f(s))$ for $s \in [s_i, s_{i+1}]$; the pasting lemma gives continuity.

Uniqueness: if $\tilde{f}'$ is another lifting and $\tilde{f}' = \tilde{f}$ on $[0, s_i]$, then $\tilde{f}'([s_i, s_{i+1}])$ is connected and lies in the disjoint union of the open slices, hence in one slice, which must be V_0 since it contains $\tilde{f}'(s_i) = \tilde{f}(s_i)$. Then $\tilde{f}'(s) = (p|V_0)^{-1}(f(s)) = \tilde{f}(s)$ there. ∎

Lemma 9.13 (54.2)

Let $p : E \to B$ be a covering map with $p(e_0) = b_0$, and $F : I \times I \to B$ continuous with $F(0,0) = b_0$. There is a unique lifting $\tilde{F} : I \times I \to E$ with $\tilde{F}(0,0) = e_0$. If F is a path homotopy, so is $\tilde{F}$.

Proof. Existence: choose subdivisions $s_0 < \cdots < s_m$ and $t_0 < \cdots < t_n$ of I such that each rectangle $I_i \times J_j = [s_{i-1}, s_i] \times [t_{j-1}, t_j]$ is mapped by F into an evenly covered open set (Lebesgue number lemma). Define $\tilde{F}$ on $A = 0 \times I \cup I \times 0$ by lifting the two edges as paths from $(0,0)$ (Lemma 9.12). Then extend over the rectangles in the order $I_1 \times J_1, \ldots, I_m \times J_1, I_1 \times J_2, \ldots$. At each stage let A' be A together with the rectangles already treated, R the next rectangle and $C = R \cap A'$, the union of the left and bottom edges of R, which is connected. With $F(R) \subset U$ evenly covered, $\tilde{F}(C)$ is connected, so lies in one slice V_0; define $\tilde{F} = (p|V_0)^{-1} \circ F$ on R. This agrees with the earlier definition on C and is continuous on $A' \cup R$ by the pasting lemma. Uniqueness: any lifting is determined on A by Lemma 9.12, and on each successive R its image is connected, contains $\tilde{F}(C) \subset V_0$, hence lies in V_0, so it equals $(p|V_0)^{-1} \circ F$. If F is a path homotopy, $F(0 \times I) = b_0$, so $\tilde{F}(0 \times I)$ is a connected subset of the discrete fibre $p^{-1}(b_0)$, hence a point; likewise for $1 \times I$. ∎

Theorem 9.14 (54.3)

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$, and f, g paths in B from b_0 to b_1 with liftings $\tilde{f}, \tilde{g}$ beginning at e_0. If $f \simeq_p g$, then $\tilde{f}$ and $\tilde{g}$ end at the same point and $\tilde{f} \simeq_p \tilde{g}$.

Proof. Let $F : f \simeq_p g$ and $\tilde{F}$ its lifting with $\tilde{F}(0,0) = e_0$, a path homotopy (Lemma 9.13). $\tilde{F}|I \times 0$ lifts f and begins at e_0, so equals $\tilde{f}$; $\tilde{F}|I \times 1$ lifts g and begins at $\tilde{F}(0,1) = e_0$, so equals $\tilde{g}$. Since $\tilde{F}(1 \times I)$ is a point, $\tilde{f}(1) = \tilde{g}(1)$. ∎

The **lifting correspondence** $\phi : \pi_1(B, b_0) \to p^{-1}(b_0)$ is $\phi([f]) = \tilde{f}(1)$, where $\tilde{f}$ lifts f from e_0; it is well defined by Theorem 9.14.

Theorem 9.15 (54.4)

If E is path connected, ϕ is surjective; if E is simply connected, ϕ is bijective.

Proof. For $e_1 \in p^{-1}(b_0)$ take a path γ in E from e_0 to e_1; γ lifts the loop $f = p \circ \gamma$, so $\phi([f]) = e_1$. If $\phi([f]) = \phi([g])$ and E is simply connected, then $\tilde{f} \simeq_p \tilde{g}$ by Lemma 9.6, say by $\tilde{F}$, and $p \circ \tilde{F}$ is a path homotopy from f to g. ∎

Theorem 9.16 (54.5)

$\pi_1(S^1, b_0)$ is isomorphic to the additive group $\mathbb{Z}$.

Proof. Use $p : \mathbb{R} \to S^1$ with $e_0 = 0$; $p^{-1}(b_0) = \mathbb{Z}$ and $\mathbb{R}$ is simply connected, so ϕ is a bijection onto $\mathbb{Z}$. If $\tilde{f}, \tilde{g}$ lift f, g from 0 with $\tilde{f}(1) = n$, $\tilde{g}(1) = m$, then $s \mapsto n+\tilde{g}(s)$ lifts g from n (as $p(n+x) = p(x)$), so $\tilde{f} * (n + \tilde{g})$ lifts $f * g$ from 0 and ends at $n + m$. Thus $\phi([f] * [g]) = \phi([f]) + \phi([g])$. ■

In particular the loop $p_0(s) = p(s)$ generates $\pi_1(S^1, b_0)$, and the class of a loop is determined by the end point n of its lifting, the number of times it winds around.

Theorem 9.17 (54.6)

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$.
(a) $p_* : \pi_1(E, e_0) \to \pi_1(B, b_0)$ is injective.
(b) Let $H = p_*(\pi_1(E, e_0))$. The lifting correspondence induces an injective map $\Phi : \pi_1(B, b_0)/H \to p^{-1}(b_0)$ of the right cosets of H, bijective if E is path connected.
(c) A loop f at b_0 has $[f] \in H$ iff f lifts to a loop at e_0.

Proof. (a) Let $\tilde{h}$ be a loop at e_0 with $p \circ \tilde{h} \simeq_p e_{b_0}$ by F. The lifting $\tilde{F}$ from e_0 is a path homotopy between the liftings of $p \circ \tilde{h}$ and e_{b_0} from e_0, namely $\tilde{h}$ and the constant loop.

(c) If $[f] = [p \circ \tilde{h}]$ with $\tilde{h}$ a loop at e_0, then by Theorem 9.14 $\tilde{f}$ ends where $\tilde{h}$ does, at e_0. Conversely, if $\tilde{f}$ is a loop, $[f] = p_*[\tilde{f}]$.

(b) Define $\Phi(H * [f]) = \phi([f])$. If $[h] \in H$, $h = p \circ \tilde{h}$ with $\tilde{h}$ a loop at e_0, then $\tilde{h} * \tilde{f}$ lifts $h * f$ and ends at $\tilde{f}(1)$; so Φ is well defined. If $\phi([f]) = \phi([g])$, then $\tilde{f} * \bar{\tilde{g}}$ is a loop at e_0 lifting $f * \bar{g}$, so $[f] * [g]^{-1} \in H$ and $H * [f] = H * [g]$. Surjectivity for path-connected E is Theorem 9.15. ■

9.5 Retractions and fixed points

Motivation. Since $\pi_1(S^1) \neq 0$ while the disc is simply connected, there is no retraction of the disc onto its boundary. This gives the Brouwer fixed-point theorem in dimension two and a criterion for when a vector field must vanish.

$A \subset X$ is a **retract** of X if there is a continuous $r : X \to A$ with $r|A = \mathrm{id}_A$ (a **retraction**).

Lemma 9.18 (55.1)

If A is a retract of X, the homomorphism j_* induced by inclusion $j : A \to X$ is injective.

Proof. $r \circ j = \mathrm{id}_A$, so $r_* \circ j_*$ is the identity. ■

Theorem 9.19 (55.2, no-retraction theorem)

There is no retraction of B^2 onto S^1.

Proof. $j_* : \pi_1(S^1) \cong \mathbb{Z} \to \pi_1(B^2) = 0$ is not injective. ■

Lemma 9.20 (55.3)

For continuous $h : S^1 \to X$ the following are equivalent:
(1) h is nulhomotopic;

(2) h extends to a continuous $k : B^2 \to X$;
(3) h_* is the trivial homomorphism.

Proof. (1)⇒(2): let $H : S^1 \times I \to X$ be a homotopy from h to a constant. The map $\pi : S^1 \times I \to B^2$, $\pi(x,t) = (1-t)x$, is continuous, closed and surjective, hence a quotient map; it collapses $S^1 \times 1$ to 0 and is otherwise injective. H is constant on $S^1 \times 1$, so it induces a continuous k with $k \circ \pi = H$, and k extends h. (2)⇒(3): $h = k \circ j$ with $j : S^1 \to B^2$, and j_* is trivial. (3)⇒(1): let $p_0(s) = p(s)$ be the standard loop and $b_0 = p_0(0)$. Then $h \circ p_0$ is path homotopic to a constant by some $F : I \times I \to X$, with $F(0,t) = F(1,t) = h(b_0)$ and $F(s,1) = h(b_0)$. The map $q : I \times I \to B^2$, $q(s,t) = (1-t)p_0(s)$, is a quotient map that identifies $(0,t)$ with $(1,t)$ and collapses $I \times 1$; F respects these identifications, so it induces $k : B^2 \to X$ with $k \circ q = F$, and k extends h. Then $H(x,t) = k((1-t)x)$ is a homotopy from h to the constant $k(0)$. ■

Corollary 9.21 (55.4)

The inclusion $j : S^1 \to \mathbb{R}^2 - 0$ and the identity $i : S^1 \to S^1$ are not nulhomotopic.

Proof. $r(x) = x/\|x\|$ retracts $\mathbb{R}^2 - 0$ onto S^1, so j_* is injective (Lemma 9.18), hence nontrivial; i_* is the identity of $\mathbb{Z}$. Apply Lemma 9.20. ■

Theorem 9.22 (55.5)

Given a nonvanishing vector field on B^2 (a continuous $v : B^2 \to \mathbb{R}^2 - 0$), there is a point of S^1 where v points directly inward and one where it points directly outward.

Proof. Suppose $v(x)$ never points directly inward, i.e. $v(x) \neq ax$ for $x \in S^1$, $a < 0$. Then $w = v|S^1 : S^1 \to \mathbb{R}^2 - 0$ extends to B^2, so is nulhomotopic. $F(x,t) = tx + (1-t)w(x)$ never vanishes on $S^1 \times I$: for $t = 0, 1$ clearly, and for $0 < t < 1$, $F = 0$ would mean $w(x) = -\frac{t}{1-t}x$. So $j \simeq w$ in $\mathbb{R}^2 - 0$ and j is nulhomotopic, contradicting Corollary 9.21. For outward, apply this to $-v$. ■

Theorem 9.23 (55.6, Brouwer fixed-point theorem for the disc)

Every continuous $f : B^2 \to B^2$ has a fixed point.

Proof. Otherwise $v(x) = f(x) - x$ is a nonvanishing vector field on B^2. At $x \in S^1$ it cannot point directly outward, since $f(x) - x = ax$ with $a > 0$ gives $f(x) = (1+a)x \notin B^2$. This contradicts Theorem 9.22. ■

Corollary 9.24 (55.7)

A 3×3 matrix A of positive real numbers has a positive real eigenvalue.

Proof. Let $B = \{x \in S^2 : x_1, x_2, x_3 \geq 0\}$; radial projection $x \mapsto x/(x_1+x_2+x_3)$ is a homeomorphism of B onto a closed triangle, which is homeomorphic to B^2. For $x \in B$, Ax has positive entries, so $x \mapsto Ax/\|Ax\|$ maps B into B. A fixed point gives $Ax = \|Ax\|x$. ■

Theorem 9.25 (55.8)

Let T be a closed triangle in $\mathbb{R}^2$. There is $\epsilon > 0$ such that for every open covering $\mathcal{A}$ of T by sets of diameter less than ϵ, some point of T belongs to at least three elements of $\mathcal{A}$.

Proof. Since no point lies on all three edges e_1, e_2, e_3, the continuous function $(a,b,c) \mapsto \max(|a-b|, |b-c|, |a-c|)$ on the compact $e_1 \times e_2 \times e_3$ is positive and has a minimum $\epsilon > 0$; so no set of diameter less than ϵ meets all three edges. Suppose $\mathcal{A}$ is such a covering in which no point lies in three elements; passing to a finite subcovering, take a dominated partition of unity $\{\phi_A\}$. For each A choose a vertex v_A such that A does not meet the edge opposite v_A. Define $k(x) = \sum_A \phi_A(x) v_A$. Each x lies in at most two elements, so $k(x)$ is a convex combination of at most two vertices and lies in ∂T. If x lies on an edge e and $\phi_A(x) > 0$, then A meets e, so e is not opposite v_A, i.e. $v_A \in e$; thus $k(e) \subset e$. Then $H(x,t) = (1-t)x + tk(x)$ is a homotopy in ∂T from the identity of ∂T to $k|\partial T$ (it stays in an edge containing x). But $k|\partial T$ extends to T, so is nulhomotopic (Lemma 9.20, $T \cong B^2$, $\partial T \cong S^1$), and then the identity of $\partial T \cong S^1$ would be nulhomotopic, contradicting Corollary 9.21. ■

So the triangle has topological dimension at least 2.

9.6 The fundamental theorem of algebra

Motivation. A topological proof: a polynomial of degree n wraps a large circle n times around the origin, and if it had no root this loop would be contractible in $\mathbb{C} - \{0\}$, contradicting $\pi_1(S^1) \cong \mathbb{Z}$.

Theorem 9.26 (56.1, fundamental theorem of algebra)

A polynomial equation $x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$ of degree $n > 0$ with real or complex coefficients has a complex root.

Proof.

Step 1. $f : S^1 \to S^1$, $f(z) = z^n$, induces an injective homomorphism: $f \circ p_0(s) = p(ns)$ lifts to $s \mapsto ns$, so f_* is multiplication by n on $\mathbb{Z}$.

Step 2. $g : S^1 \to \mathbb{R}^2 - 0$, $g(z) = z^n$, equals $j \circ f$, so $g_* = j_* f_*$ is injective and g is not nulhomotopic.

Step 3. if $\sum |a_i| < 1$, there is a root in B^2. Otherwise $k(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$ maps B^2 into $\mathbb{R}^2 - 0$, so $h = k|S^1$ is nulhomotopic. But $F(z,t) = z^n + t(a_{n-1}z^{n-1} + \cdots + a_0)$ never vanishes on $S^1 \times I$, since $|t(a_{n-1}z^{n-1} + \cdots + a_0)| \le \sum |a_i| < 1 = |z^n|$; so $g \simeq h$ in $\mathbb{R}^2 - 0$, a contradiction.

Step 4. in general, choose $c > 0$ with $\sum_i |a_{n-i}|/c^i < 1$ and substitute $x = cy$: the equation $y^n + (a_{n-1}/c)y^{n-1} + \cdots + a_0/c^n = 0$ has a root y_0, and cy_0 is a root of the original. ■

9.7 The Borsuk–Ulam theorem

Motivation. Some point on the earth has the same temperature and pressure as its antipode. The Borsuk–Ulam theorem for S^2 proves this, again using the fact that a nontrivial loop in S^1 cannot be killed.

$h : S^n \to S^m$ is **antipode-preserving** if $h(-x) = -h(x)$.

Theorem 9.27 (57.1)

A continuous antipode-preserving $h : S^1 \to S^1$ is not nulhomotopic.

Proof. Composing with a rotation (which commutes with $x \mapsto -x$) we may assume $h(b_0) = b_0$. Regard $S^1 \subset \mathbb{C}$ and let $q(z) = z^2$, a quotient map (continuous, closed, surjective) identifying z with $-z$. Since $q(h(-z)) = q(-h(z)) = q(h(z))$, there is a continuous $k : S^1 \to S^1$ with $k \circ q = q \circ h$.

Now q_* is injective ($q\circ p_0$ lifts to $s\mapsto 2s$, so q_* is multiplication by 2). Let $f(s)=e^{\pi i s}$, a path from b_0 to $-b_0$; then $q\circ f=p_0$. The path $h\circ f$ runs from b_0 to $-b_0$, so its lifting $\tilde g$ from 0 ends at some $m+\frac12$; $q\circ h\circ f=p\circ 2\tilde g$, and $2\tilde g$ ends at the odd integer $2m+1$. So $k_*[p_0]=[k\circ q\circ f]=[q\circ h\circ f]$ corresponds to $2m+1\neq 0$, and k_* is injective. If h were nulhomotopic, h_* would be trivial, hence $k_*q_*=q_*h_*$ trivial, contradicting the injectivity of k_*q_*. ■

Theorem 9.28 (57.2)

There is no continuous antipode-preserving map $g:S^2\to S^1$.

Proof. Its restriction h to the equator S^1 is antipode-preserving, hence not nulhomotopic. But h extends to the upper hemisphere, which is homeomorphic to B^2 by $(x,y,z)\mapsto(x,y)$; by Lemma 9.20 h is nulhomotopic. ■

Theorem 9.29 (57.3, Borsuk–Ulam theorem for S^2)

For every continuous $f:S^2\to\mathbb{R}^2$ there is $x\in S^2$ with $f(x)=f(-x)$.

Proof. Otherwise $g(x)=(f(x)-f(-x))/\|f(x)-f(-x)\|$ is a continuous antipode-preserving map $S^2\to S^1$. ■

Theorem 9.30 (57.4, bisection theorem)

Given two bounded polygonal regions in $\mathbb{R}^2$, some line bisects each of them.

Proof. Put the regions A_1,A_2 in the plane $z=1$ of $\mathbb{R}^3$. For $u\in S^2$, let $f_i(u)$ be the area of the part of A_i on the side $x\cdot u>0$ of the plane $P_u=\{x\cdot u=0\}$. Then $f_i(-u)=\operatorname{area}(A_i)-f_i(u)$, and f_i is continuous (we assume this elementary fact about areas). Borsuk–Ulam applied to (f_1,f_2) gives u with $f_i(u)=f_i(-u)=\frac12\operatorname{area}(A_i)$. Here $u\neq\pm(0,0,1)$, since then $f_i(u)$ is 0 or the full area; so P_u meets the plane $z=1$ in a line, which bisects both regions. ■

9.8 Deformation retracts and homotopy type

Motivation. The punctured plane and the circle have the same fundamental group because one can be squeezed onto the other. Homotopy equivalence makes this idea precise and lets us replace a space by a simpler one before computing.

Lemma 9.31 (58.1)

If $h,k:(X,x_0)\to(Y,y_0)$ are homotopic by a homotopy keeping x_0 at y_0, then $h_*=k_*$.

Proof. For a loop f at x_0, $G(s,t)=H(f(s),t)$ is a path homotopy from $h\circ f$ to $k\circ f$. ■

Theorem 9.32 (58.2)

The inclusion $j:S^n\to\mathbb{R}^{n+1}-0$ induces an isomorphism of fundamental groups.

Proof. $r(x)=x/\|x\|$ satisfies $r\circ j=\mathrm{id}$, and $H(x,t)=(1-t)x+tx/\|x\|$ is a homotopy in $\mathbb{R}^{n+1}-0$ from the identity to $j\circ r$ fixing S^n pointwise. By Lemma 9.31, $j_*r_*=\mathrm{id}$, and $r_*j_*=\mathrm{id}$. ■

$A \subset X$ is a **deformation retract** of X if the identity of X is homotopic to a map carrying X into A by a homotopy fixing each point of A: $H : X \times I \to X$ with $H(x,0) = x$, $H(x,1) \in A$, $H(a,t) = a$ for $a \in A$.

Theorem 9.33 (58.3)

If A is a deformation retract of X and $x_0 \in A$, then $j : (A, x_0) \to (X, x_0)$ induces an isomorphism.

Proof. As in Theorem 9.32, with $r(x) = H(x,1)$. ∎

Example 9.4

The doubly punctured plane $\mathbb{R}^2 - \{p, q\}$ deformation retracts onto a figure eight, and also onto a theta space; the solid torus $B^2 \times S^1$ and $\mathbb{R}^3$ minus the z-axis deformation retract onto a circle, so their fundamental groups are $\mathbb{Z}$.

$f : X \to Y$ is a **homotopy equivalence** with **homotopy inverse** $g : Y \to X$ if $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$; then X, Y have the same **homotopy type**.

Lemma 9.34 (58.4)

Let $h, k : X \to Y$ be homotopic by H, with $h(x_0) = y_0$, $k(x_0) = y_1$, and let $\alpha(t) = H(x_0, t)$. Then $k_* = \hat{\alpha} \circ h_*$.

Proof. Let f be a loop at x_0 and $G(s,t) = H(f(s), t)$ on $I \times I$. Consider the paths $\beta_0(s) = (s,0)$, $\beta_1(s) = (s,1)$, $\gamma_0(t) = (0,t)$, $\gamma_1(t) = (1,t)$ in $I \times I$. Both $\beta_0 * \gamma_1$ and $\gamma_0 * \beta_1$ run from $(0,0)$ to $(1,1)$ in the convex set $I \times I$, hence are path homotopic, and so are their images under G: $(h \circ f) * \alpha \simeq_p \alpha * (k \circ f)$. Thus $[k \circ f] = [\bar{\alpha}] * [h \circ f] * [\alpha]$. ∎

Corollary 9.35 (58.5)

If $h \simeq k$ and h_* is injective, surjective or trivial, so is k_*.

Corollary 9.36 (58.6)

If h is nulhomotopic, h_* is trivial.

Proof. Both follow from Lemma 9.34, since $\hat{\alpha}$ is an isomorphism and a constant map induces the trivial homomorphism. ∎

Theorem 9.37 (58.7)

If $f : X \to Y$ is a homotopy equivalence with $f(x_0) = y_0$, then $f_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ is an isomorphism.

Proof. Let g be a homotopy inverse, $x_1 = g(y_0)$, $y_1 = f(x_1)$. Since $g \circ f \simeq \mathrm{id}_X$, Lemma 9.34 shows that $g_* \circ (f_{x_0})_* : \pi_1(X, x_0) \to \pi_1(X, x_1)$ is an isomorphism, so $(g_{y_0})_*$ is surjective. Since $f \circ g \simeq \mathrm{id}_Y$, $(f_{x_1})_* \circ (g_{y_0})_* : \pi_1(Y, y_0) \to \pi_1(Y, y_1)$ is an isomorphism, so $(g_{y_0})_*$ is injective. Hence $(g_{y_0})_*$ is an isomorphism, and so is $(f_{x_0})_*$. ∎

9.9 The fundamental group of S^n

Motivation. Higher spheres are simply connected. The proof introduces the idea, made systematic by van Kampen's theorem in Chapter 11, of computing π_1 from a cover by open sets with known fundamental groups.

Theorem 9.38 (59.1)

Let $X = U \cup V$ with U, V open, $U \cap V$ path connected and $x_0 \in U \cap V$. Then the images of $i_* : \pi_1(U, x_0) \to \pi_1(X, x_0)$ and $j_* : \pi_1(V, x_0) \to \pi_1(X, x_0)$ generate $\pi_1(X, x_0)$.

Proof. Let f be a loop at x_0. By the Lebesgue number lemma choose $0 = a_0 < \cdots < a_n = 1$ with each $f([a_{i-1}, a_i])$ in U or in V. If some $f(a_i)$ $(0 < i < n)$ is not in $U \cap V$, say it lies in $U - V$, then both adjacent intervals are mapped into U, and a_i can be deleted; repeating, we may assume $f(a_i) \in U \cap V$ for all i. Choose paths α_i in $U \cap V$ from x_0 to $f(a_i)$ (α_0, α_n constant). With f_i as in Theorem 9.3, $[f] = [f_1] * \cdots * [f_n] = [g_1] * \cdots * [g_n]$ where $g_i = \alpha_{i-1} * f_i * \bar{\alpha}_i$ is a loop at x_0 lying entirely in U or in V. ∎

Corollary 9.39 (59.2)

If $X = U \cup V$ with U, V open and simply connected and $U \cap V$ nonempty and path connected, then X is simply connected.

Theorem 9.40 (59.3)

For $n \geq 2$, S^n is simply connected.

Proof. Let $p = (0, \ldots, 0, 1)$. Stereographic projection $f(x) = (x_1, \ldots, x_n)/(1 - x_{n+1})$ is a homeomorphism $S^n - p \to \mathbb{R}^n$, with inverse $y \mapsto (t(y)y, 1 - t(y))$, $t(y) = 2/(1 + \|y\|^2)$; similarly $S^n - (-p) \cong \mathbb{R}^n$. Both are simply connected, and their intersection is homeomorphic to $\mathbb{R}^n - 0$, which is path connected for $n \geq 2$. Apply Corollary 9.39. ∎

9.10 Fundamental groups of some surfaces

Motivation. The torus, the projective plane and the double torus can now be distinguished from each other and from the sphere by their fundamental groups, giving the first steps toward the classification of surfaces.

A **surface** is a Hausdorff space with a countable basis in which each point has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^2$.

Theorem 9.41 (60.1)

$\pi_1(X \times Y, x_0 \times y_0) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$.

Proof. With projections p, q, let $\Phi([f]) = (p_*[f], q_*[f])$, a homomorphism. Given loops g in X and h in Y, $f(s) = g(s) \times h(s)$ maps to $([g], [h])$. If $p \circ f \simeq_p e$ by G and $q \circ f \simeq_p e$ by H, then $G \times H$ is a path homotopy from f to the constant loop. ∎

Corollary 9.42 (60.2)

The fundamental group of the torus $S^1 \times S^1$ is $\mathbb{Z} \times \mathbb{Z}$.

The **projective plane** P^2 is the quotient of S^2 obtained by identifying each x with $-x$.

Theorem 9.43 (60.3)

P^2 is a compact surface, and the quotient map $p : S^2 \to P^2$ is a covering map.

Proof. p is open: for U open, $p^{-1}(p(U)) = U \cup a(U)$ with $a(x) = -x$, which is open. For $x \in S^2$ let U be the ϵ-neighbourhood of x with $\epsilon < 1$; it contains no antipodal pair, so $p|U$ is a continuous open bijection onto $p(U)$, a homeomorphism, and similarly for $a(U)$, which is disjoint from U. So $p(U)$ is evenly covered with slices $U, a(U)$. Hausdorff: for $p(x) \neq p(y)$, the four points $\pm x, \pm y$ are distinct; take ϵ less than half their minimum distance, so the ϵ-neighbourhoods U of x and W of y satisfy $(U \cup a(U)) \cap (W \cup a(W)) = \emptyset$, whence $p(U) \cap p(W) = \emptyset$. A countable basis of S^2 maps to one of P^2 since p is open. Locally, P^2 is homeomorphic to open subsets of S^2, hence of $\mathbb{R}^2$; and it is compact as a continuous image of S^2. ■

Corollary 9.44 (60.4)

$\pi_1(P^2, y)$ has order 2.

Proof. S^2 is simply connected, so the lifting correspondence is a bijection onto the two-point fibre (Theorem 9.15). ■

Similarly P^n $(n \geq 2)$ has fundamental group of order 2.

Lemma 9.45 (60.5)

The fundamental group of the figure eight is not abelian.

Proof. Let $X = A \cup B$ be two circles meeting at x_0. Let E be the union of the coordinate axes in $\mathbb{R}^2$ together with, for each nonzero integer n, a circle tangent to the x-axis at $(n, 0)$ and a circle tangent to the y-axis at $(0, n)$. Define $p : E \to X$ by wrapping the x-axis around A (each $[n, n+1]$ once, integers to x_0), the y-axis around B, each circle tangent to the x-axis homeomorphically onto B and each circle tangent to the y-axis onto A (tangency points to x_0). Points of $X - x_0$ have small arc neighbourhoods whose preimages are disjoint arcs mapped homeomorphically, and x_0 has a small "cross" neighbourhood whose preimage is a disjoint union of crosses about the integer lattice points on the axes; so p is a covering map. Let g and h be loops going once around A and B. The lifting of $g * h$ from the origin runs along the x-axis to $(1, 0)$ and then around the circle tangent there, ending at $(1, 0)$; the lifting of $h * g$ ends at $(0, 1)$. By Theorem 9.14, $g * h \not\simeq_p h * g$. ■

Theorem 9.46 (60.6)

The fundamental group of the double torus is not abelian.

Proof. The double torus X contains a figure eight A, consisting of one circle around each handle meeting at a point, and A is a retract of X: each handle can be squashed continuously onto its circle, compatibly along the curve where the two halves are joined. By Lemma 9.18, $\pi_1(X)$ contains a copy of the nonabelian group $\pi_1(A)$. ■

Corollary 9.47 (60.7)

The 2-sphere, torus, projective plane and double torus are topologically distinct.

Proof. Their fundamental groups are trivial, $\mathbb{Z} \times \mathbb{Z}$ (infinite abelian), of order 2, and nonabelian. ■

Chapter 10. Separation Theorems in the Plane

10.1 The Jordan separation theorem

Motivation. Every simple closed curve in the plane separates it into an inside and an outside. The statement is obvious for circles and polygons but surprisingly hard for wild curves; fundamental groups provide the proof. We begin with the easier half, that the complement is disconnected.

Throughout, open subsets of S^2 and $\mathbb{R}^2$ are locally path connected, so their components are open and coincide with their path components. A **separates** a connected space X if $X - A$ is not connected, and separates it into n components if $X - A$ has n components. An **arc** is a space homeomorphic to $[0,1]$; a **simple closed curve** is a space homeomorphic to S^1.

Lemma 10.1 (61.1)

Let $C \subset S^2$ be compact, $b \in S^2 - C$, and $h : S^2 - b \to \mathbb{R}^2$ a homeomorphism. If U is a component of $S^2 - C$ not containing b, then $h(U)$ is a bounded component of $\mathbb{R}^2 - h(C)$; if U contains b, then $h(U - b)$ is the unbounded component. In particular $S^2 - C$ and $\mathbb{R}^2 - h(C)$ have the same number of components.

Proof. First, $U - b$ is connected for every component U. Suppose $b \in U$ and A, B separate $U - b$. Choose a neighbourhood W of b disjoint from C and homeomorphic to an open disc; $W \subset U$, and the connected set $W - b$ lies in A, say. Then b is not a limit point of B, and $A \cup \{b\}$, B separate U, a contradiction. Hence the sets $h(U_\alpha - b)$, for the components U_α of $S^2 - C$, are disjoint connected open sets with union $\mathbb{R}^2 - h(C)$; they are its components. Extend h to a homeomorphism H of S^2 with the one-point compactification $\mathbb{R}^2 \cup \{\infty\}$ by $H(b) = \infty$. For the component $U \ni b$, $H(U)$ is a neighbourhood of ∞, so $h(U - b)$ contains the complement of a compact set: it is unbounded, and all the other components lie in a compact set, hence are bounded. ∎

Lemma 10.2 (61.2, nulhomotopy lemma)

Let $a, b \in S^2$, A compact and $f : A \to S^2 - a - b$ continuous. If a and b lie in the same component of $S^2 - f(A)$, then f is nulhomotopic.

Proof. Identify S^2 with $\mathbb{R}^2 \cup \{\infty\}$, a with 0 and b with ∞. We must show: if $g : A \to \mathbb{R}^2 - 0$ and 0 lies in the unbounded component of $\mathbb{R}^2 - g(A)$, then g is nulhomotopic. Choose a ball B about 0 containing $g(A)$ and a point p outside B; 0 and p lie in the unbounded component, so there is a path α in $\mathbb{R}^2 - g(A)$ from 0 to p. Then $G(x,t) = g(x) - \alpha(t)$ is a homotopy in $\mathbb{R}^2 - 0$ from g to $k(x) = g(x) - p$, and $H(x,t) = tg(x) - p$ is a homotopy in $\mathbb{R}^2 - 0$ from the constant $-p$ to k (since $tg(x) \in B$ and $p \notin B$). ∎

Theorem 10.3 (61.3, Jordan separation theorem)

A simple closed curve C in S^2 separates S^2.

Proof. Suppose $S^2 - C$ is connected, hence path connected. Write $C = A_1 \cup A_2$ with arcs meeting only in their end points a, b, and let $X = S^2 - a - b$, $U = S^2 - A_1$, $V = S^2 - A_2$. Then $X = U \cup V$ and $U \cap V = S^2 - C$ is path connected. We show $i_* : \pi_1(U, x_0) \to \pi_1(X, x_0)$ is trivial; similarly j_*; then $\pi_1(X, x_0)$ is trivial by Theorem 9.38, whereas $X \cong \mathbb{R}^2 - 0$. Let f be a loop in U at x_0; it induces $h : S^1 \to U$ with $h \circ p_0 = f$. The set $h(S^1)$ misses the connected set $A_1 \ni a, b$, so a and b lie in one component of $S^2 - h(S^1)$, and $i \circ h$ is nulhomotopic (Lemma 10.2). By Lemma 9.20, $(i \circ h)_*$ is trivial, and $i_*[f] = (i \circ h)_*[p_0]$. ∎

Theorem 10.4 (61.4, general separation theorem)

If A_1, A_2 are closed connected subsets of S^2 with $A_1 \cap A_2 = \{a, b\}$, then $C = A_1 \cup A_2$ separates S^2.

Proof. $C \neq S^2$, since $S^2 - a - b$ is connected while $C - a - b$ is separated by the sets $A_i - a - b$. The rest of the proof of Theorem 10.3 used only that A_1, A_2 are closed and connected and meet in $\{a, b\}$. ■

10.2 Invariance of domain

Motivation. If $U \subseteq \mathbb{R}^2$ is open and $f : U \to \mathbb{R}^2$ is continuous and injective, then $f(U)$ is open. This is the two-dimensional case of Brouwer's theorem and is needed in the Jordan curve theorem and in the theory of manifolds.

Lemma 10.5 (62.1, homotopy extension lemma)

Let $X \times I$ be normal, $A \subset X$ closed, and $f : A \to Y$ continuous with Y open in $\mathbb{R}^n$. If f is nulhomotopic, it extends to a continuous nulhomotopic $g : X \to Y$.

Proof. Let $F : A \times I \to Y$ be a homotopy from f to the constant y_0, and set $F(x, 1) = y_0$ for $x \in X$. This is continuous on the closed set $(A \times I) \cup (X \times 1)$, so by Tietze (coordinatewise) it extends to $G : X \times I \to \mathbb{R}^n$. The open set $U = G^{-1}(Y)$ contains $(A \times I) \cup (X \times 1)$; by the tube lemma each $a \in A$ has a neighbourhood W_a with $W_a \times I \subset U$, and $W = \bigcup W_a$ is open with $A \subset W$, $W \times I \subset U$. $X \cong X \times 0$ is normal, so Urysohn gives $\phi : X \to [0, 1]$ with $\phi = 0$ on A, $\phi = 1$ off W. Then $(x, \phi(x)) \in U$ always, and $g(x) = G(x, \phi(x))$ maps X into Y and extends f. The homotopy $H(x, t) = G(x, (1 - t)\phi(x) + t)$ stays in Y for the same reason and joins g to the constant y_0. ■

Lemma 10.6 (62.2, Borsuk lemma)

Let $a, b \in S^2$, A compact, and $f : A \to S^2 - a - b$ continuous and injective. If f is nulhomotopic, then a and b lie in the same component of $S^2 - f(A)$.

Proof. $f(A)$ is homeomorphic to A, so we may take f to be an inclusion; identify a with 0 and b with ∞. We show: if $A \subset \mathbb{R}^2 - 0$ is compact and the inclusion $j : A \to \mathbb{R}^2 - 0$ is nulhomotopic, then 0 lies in the unbounded component of $\mathbb{R}^2 - A$. Suppose the component C containing 0 is bounded, and let D be the union of the other components; $\mathbb{R}^2 - A = C \cup D$, both open. $C \cup A = \mathbb{R}^2 - D$ is closed and bounded, hence compact, and Lemma 10.5 extends j to $k : C \cup A \to \mathbb{R}^2 - 0$. Define $h = k$ on $C \cup A$ and $h(x) = x$ on $D \cup A$; by the pasting lemma $h : \mathbb{R}^2 \to \mathbb{R}^2 - 0$ is continuous. Choose a closed ball B of radius M whose interior contains $C \cup A$. Then $h|B$ is the identity on $\operatorname{Bd} B$, and $x \mapsto Mh(x)/\|h(x)\|$ is a retraction of B onto $\operatorname{Bd} B$, contradicting Theorem 9.19. ■

Theorem 10.7 (62.3, invariance of domain)

If $U \subset \mathbb{R}^2$ is open and $f : U \to \mathbb{R}^2$ is continuous and injective, then $f(U)$ is open and $f^{-1} : f(U) \to U$ is continuous.

Proof. Regard f as a map into $S^2 \supset \mathbb{R}^2$; it suffices that f is an open map into S^2.

Step 1. for a closed ball $B \subset U$, $f(B)$ does not separate S^2. If $a, b \notin f(B)$, then $f|B : B \to S^2 - a - b$ is nulhomotopic (as B is contractible) and injective, so a, b lie in one component of $S^2 - f(B)$ (Lemma 10.6).

Step 2. $f(\operatorname{Int} B)$ is open. $C = f(\operatorname{Bd} B)$ is a simple closed curve, so it separates S^2 (Theorem 10.3). Let V be the component of $S^2 - C$ containing the connected set $f(\operatorname{Int} B)$ and $W \neq \emptyset$ the union of the others. If $a \in V - f(\operatorname{Int} B)$, then $a \notin f(B)$; pick $b \in W$. By Step 1, a, b lie in a connected subset of $S^2 - f(B) \subset S^2 - C$, so in the same component of $S^2 - C$, a contradiction. Hence $f(\operatorname{Int} B) = V$ is open.

Step 3. every open subset of U is a union of interiors of closed balls in U, so f is an open map. ■

The theorem holds in $\mathbb{R}^n$ for all n, but the proof requires higher homotopy or homology groups.

10.3 The Jordan curve theorem

Motivation. We complete the proof: the complement of a simple closed curve has exactly two components, and the curve is the boundary of each.

Theorem 10.8 (63.1)

Let $X = U \cup V$ with U, V open and $U \cap V = A \cup B$, A, B disjoint and open. Let α be a path in U from $a \in A$ to $b \in B$, β a path in V from b to a, and $f = \alpha * \beta$.
(a) $[f]$ generates an infinite cyclic subgroup of $\pi_1(X, a)$.
(b) If $\pi_1(X, a)$ is infinite cyclic, it is generated by $[f]$.
(c) If γ is a path in U from a to $a' \in A$ and δ a path in V from a' to a, and $g = \gamma * \delta$, then the subgroups generated by $[f]$ and $[g]$ meet only in the identity.

Proof.

Step 1 (a covering space). Let Y be the disjoint union of copies $U \times 2n$ and $V \times (2n+1)$, $n \in \mathbb{Z}$, inside $X \times \mathbb{Z}$, and let E be the quotient of Y obtained by identifying $x \times 2n$ with $x \times (2n+1)$ for $x \in B$ and $x \times 2n$ with $x \times (2n-1)$ for $x \in A$. Let $\pi : Y \to E$ be the quotient map; $x \times m \mapsto x$ induces a continuous surjection $p : E \to X$. π is open: for W open in U, $\pi^{-1}\pi(W \times 2n) = W \times 2n \cup (W \cap B) \times (2n+1) \cup (W \cap A) \times (2n-1)$, which is open; similarly for $V \times (2n+1)$. Hence $p^{-1}(U)$ is the disjoint union of the open sets $\pi(U \times 2n)$, and $\pi | U \times 2n$ is a continuous open bijection onto its image, so p maps each $\pi(U \times 2n)$ homeomorphically onto U; likewise for V. So p is a covering map.

Step 2. Let $e_n = \pi(a \times 2n)$; these are the distinct points of $p^{-1}(a)$. The paths $\tilde{\alpha}_n(s) = \pi(\alpha(s) \times 2n)$ and $\tilde{\beta}_n(s) = \pi(\beta(s) \times (2n+1))$ lift α, β; $\tilde{\alpha}_n$ ends at $\pi(b \times 2n) = \pi(b \times (2n+1))$, where $\tilde{\beta}_n$ begins, and $\tilde{\beta}_n$ ends at $\pi(a \times (2n+1)) = e_{n+1}$. So $\tilde{f}_n = \tilde{\alpha}_n * \tilde{\beta}_n$ lifts f from e_n to e_{n+1}.

Step 3 (a). For $m > 0$, $\tilde{f}_0 * \cdots * \tilde{f}_{m-1}$ lifts f^m from e_0 to $e_m \neq e_0$, so $[f]^m \neq 1$.

Step 4 (b). The lifting correspondence ϕ sends $[f]^m$ to e_m for all $m \in \mathbb{Z}$ (similarly for negative m), so it is surjective. By Theorem 9.17, ϕ induces a bijection $\pi_1(X, a)/H \to p^{-1}(a)$, $H = p_*\pi_1(E, e_0)$; the fibre is infinite while a quotient of $\mathbb{Z}$ by a nontrivial subgroup is finite, so H is trivial and ϕ is bijective. Since ϕ maps the subgroup generated by $[f]$ onto the fibre, this subgroup is all of $\pi_1(X, a)$.

Step 5 (c). $\tilde{\gamma}(s) = \pi(\gamma(s) \times 0)$ and $\tilde{\delta}(s) = \pi(\delta(s) \times (-1))$ lift γ, δ, and $\pi(a' \times 0) = \pi(a' \times (-1))$ as $a' \in A$; $\tilde{\delta}$ ends at $\pi(a \times (-1)) = e_0$. So g lifts to a loop at e_0, and so does every power of g, while f^m $(m \neq 0)$ lifts to a path ending at $e_m \neq e_0$. Hence $[f]^m \neq [g]^k$ unless $m = 0$, which forces $[g]^k = 1$. ■

Theorem 10.9 (63.2, nonseparation theorem)

An arc D in S^2 does not separate S^2.

Proof. D is contractible, so for $a, b \notin D$ the inclusion $D \to S^2 - a - b$ is nulhomotopic; by Lemma 10.6, a and b lie in the same component of $S^2 - D$. ∎

A second proof gives the following generalization.

Theorem 10.10 (63.3, general nonseparation theorem)

Let D_1, D_2 be closed subsets of S^2 with $S^2 - (D_1 \cap D_2)$ simply connected. If neither D_1 nor D_2 separates S^2, neither does $D_1 \cup D_2$.

Proof. Let $a, b \notin D_1 \cup D_2$ and suppose they cannot be joined by a path in $S^2 - (D_1 \cup D_2)$. Let $X = S^2 - (D_1 \cap D_2)$, $U = S^2 - D_1$, $V = S^2 - D_2$; then $X = U \cup V$ and $U \cap V = S^2 - (D_1 \cup D_2)$. Let A be the path component of $U \cap V$ containing a and B the union of the others, both open. Since D_1, D_2 do not separate, there are paths in U from a to b and in V from b to a. Theorem 10.8(a) shows $\pi_1(X, a)$ is nontrivial, a contradiction. ∎

(For an arc $D = h([0,1])$, this yields the second proof of Theorem 10.9: if a, b cannot be joined in $S^2 - D$, they cannot be joined in $S^2 - h(I_1)$ for one of the halves I_1 of $[0,1]$, since S^2−point is simply connected; bisecting repeatedly gives nested I_n shrinking to a point x, but a path from a to b in $S^2 - h(x)$ has compact image missing $h(I_n)$ for large n.)

Theorem 10.11 (63.4, Jordan curve theorem)

A simple closed curve C in S^2 separates S^2 into precisely two components W_1, W_2, and $C = \overline{W_i} - W_i$ for $i = 1, 2$.

Proof.

Step 1. write $C = C_1 \cup C_2$ with arcs meeting in $\{p, q\}$; let $X = S^2 - p - q$, $U = S^2 - C_1$, $V = S^2 - C_2$, so $U \cap V = S^2 - C$, which has at least two components (Theorem 10.3). Suppose it has more: let A_1, A_2 be two components and B the union of the others, all open; pick $a \in A_1$, $a' \in A_2$, $b \in B$. The arcs C_i do not separate S^2 (Theorem 10.9), so there are paths α, γ in U from a to b and from a to a', and β, δ in V from b to a and from a' to a. With $f = \alpha * \beta$ and $g = \gamma * \delta$, Theorem 10.8(a) applied to the decomposition $U \cap V = (A_1 \cup A_2) \cup B$ shows $[f] \neq 1$ has infinite order, and applied to $A_1 \cup (A_2 \cup B)$ shows the same for $[g]$. Since $\pi_1(X, a) \cong \mathbb{Z}$, $[f]^m = [g]^k$ for some nonzero m, k, contradicting Theorem 10.8(c) for the decomposition $(A_1 \cup A_2) \cup B$, in which $a, a' \in A_1 \cup A_2$.

Step 2. W_1, W_2 are open, so neither contains a limit point of the other, and $\overline{W_i} - W_i \subset C$. Conversely let $x \in C$ and U a neighbourhood of x. Write $C = C_1 \cup C_2$ with arcs meeting in their end points and $C_1 \subset U$. For $a \in W_1$, $b \in W_2$, take a path α from a to b in $S^2 - C_2$ (Theorem 10.9). Its connected image meets the disjoint open sets W_1 and $S^2 - \overline{W_1} \supset W_2$, so it is not contained in their union and contains a point y of $\overline{W_1} - W_1 \subset C$; as it misses C_2, $y \in C_1 \subset U$. So $x \in \overline{W_1} - W_1$, and likewise for W_2. ∎

Theorem 10.12 (63.5)

If C_1, C_2 are closed connected subsets of S^2 meeting in exactly two points, and neither separates S^2, then $C_1 \cup C_2$ separates S^2 into precisely two components.

Proof. At least two components by Theorem 10.4; at most two by Step 1 above, which used only these properties. ∎

The union of two arcs meeting in two points that are not both end points separates S^2 into two components, but the union need not be their common boundary.

10.4 Imbedding graphs in the plane

Motivation. Can three houses be connected to three utilities without crossing lines? The Jordan curve theorem shows that the graph $K_{3,3}$ cannot be embedded in the plane.

A **linear graph** is a Hausdorff space that is a finite union of arcs (**edges**), any two meeting in at most a common end point (a **vertex**). The **complete graph on n vertices** has an edge joining each pair of vertices; the **utilities graph** has vertices g, w, e, h_1, h_2, h_3 with an edge joining each of g, w, e to each h_i. A **theta space** is a Hausdorff space that is the union of three arcs A, B, C, each pair meeting precisely in their common end points.

Lemma 10.13 (64.1)

A theta space $X = A \cup B \cup C$ in S^2 separates S^2 into three components, with boundaries $A \cup B$, $B \cup C$, $A \cup C$. The component with boundary $A \cup B$ is a component of $S^2 - (A \cup B)$.

Proof. Let a, b be the common end points. By the Jordan curve theorem, $A \cup B$ separates S^2 into U and U' with boundary $A \cup B$. The connected set $C - a - b$ lies in one, say U'. The connected sets $\overline{U} = U \cup A \cup B$ and C meet only in $\{a, b\}$, and neither separates S^2 (C is an arc, and $S^2 - \overline{U} = U'$ is connected). By Theorem 10.12, $\overline{U} \cup C$ separates S^2 into two components V, W. So $S^2 - X = U \cup V \cup W$ is a union of three disjoint open connected sets, which are its components; U has boundary $A \cup B$ and is a component of $S^2 - (A \cup B)$. By symmetry the other two components have boundaries $B \cup C$ and $A \cup C$. ∎

Theorem 10.14 (64.2)

The utilities graph cannot be imbedded in the plane.

Proof. Suppose it lies in S^2. The arcs $A = gh_1 \cup h_1 w$, $B = gh_2 \cup h_2 w$, $C = gh_3 \cup h_3 w$ form a theta space, which separates S^2 into U, V, W with boundaries $A \cup B$, $B \cup C$, $A \cup C$. The vertex e lies in one of them, and the connected sets $eh_i - h_i$ miss $A \cup B \cup C$, so lie in that component; then h_1, h_2, h_3 lie in its boundary. But $A \cup B$ misses h_3, $B \cup C$ misses h_1, $A \cup C$ misses h_2. ∎

Lemma 10.15 (64.3)

A complete graph X on four vertices $a_1, \ldots, a_4$ in S^2 separates S^2 into four components, whose boundaries are the subgraphs X_i consisting of the edges not having a_i as a vertex.

Proof. Let Y be the union of all edges except a_2a_4; it is the theta space with arcs $A = a_1a_2 \cup a_2a_3$, $B = a_1a_3$, $C = a_1a_4 \cup a_4a_3$. It separates S^2 into U, V, W with boundaries $A \cup B$, $B \cup C$, $A \cup C$. The connected set $a_2a_4 - a_2 - a_4$ lies in one of them, whose boundary contains a_2 and a_4: not U ($a_4 \notin A \cup B$), not V ($a_2 \notin B \cup C$), so W. Now $\overline{U} \cup \overline{V}$ is connected (both contain B), and does not separate S^2, its complement being W; the arc a_2a_4 does not separate S^2; and they meet only in a_2, a_4. By Theorem 10.12 their union separates S^2 into W_1, W_2. Thus $S^2 - X = U \cup V \cup W_1 \cup W_2$, and these open connected sets are its components. U has boundary $A \cup B = X_4$; by symmetry the others have boundaries X_1, X_2, X_3. ∎

Theorem 10.16 (64.4)

The complete graph on five vertices cannot be imbedded in the plane.

Proof. Suppose $G \subset S^2$ has vertices $a_1, \ldots, a_5$, and let X be the complete subgraph on $a_1, \ldots, a_4$. The vertex a_5 lies in one of the four components of $S^2 - X$, and so does the connected set formed

by the edges at a_5 minus $a_1, \dots, a_4$. Hence $a_1, \dots, a_4$ all lie in the boundary of that component, which is some X_i; but X_i does not contain a_i. ■

By a theorem of Kuratowski, a graph imbeds in the plane iff it contains no subgraph homeomorphic to the utilities graph or the complete graph on five vertices.

10.5 Winding number of simple closed curves

Motivation. How many times does a curve wind around a point? The winding number, defined through the fundamental group of the punctured plane, is ± 1 inside a simple closed curve and 0 outside.

Lemma 10.17 (65.1)

Let $G \subset S^2$ be a complete graph on $a_1, \dots, a_4$, C the simple closed curve $a_1a_2a_3a_4a_1$, and p, q interior points of the edges a_1a_3 and a_2a_4. Then (a) p and q lie in different components of $S^2 - C$;
(b) the inclusion $j : C \to S^2 - p - q$ induces an isomorphism of fundamental groups.

Proof. (a) The theta space $C \cup a_1a_3$ (arcs $a_1a_2a_3$, a_1a_3, $a_1a_4a_3$) separates S^2 into three components; one of them, W, has boundary C, and it is the only one whose boundary contains both a_2 and a_4. So the connected set $a_2a_4 - a_2 - a_4$ lies in W and $q \in W$. By Lemma 10.13, W is a component of $S^2 - C$, and $p \notin W$ since p lies on the theta space. So p, q lie in different components of $S^2 - C$.

(b) Let $X = S^2 - p - q$. Choose x interior to a_1a_2 and y interior to a_3a_4, and let α be the path in C from x through a_2, a_3 to y and β the path in C from y through a_4, a_1 to x. Let D_1 be the arc of the simple closed curve $D = a_1a_3a_2a_4a_1$ from p through a_1, a_4 to q, and D_2 the arc from p through a_3, a_2 to q; put $U = S^2 - D_1$, $V = S^2 - D_2$. Then $X = U \cup V$, $U \cap V = S^2 - D$ has two components (Jordan curve theorem), and x, y lie in different ones by (a) applied to G with the curve D (whose "diagonals" are a_1a_2 and a_3a_4). Also $\alpha \subset U$ and $\beta \subset V$. As $\pi_1(X, x) \cong \mathbb{Z}$, Theorem 10.8(b) shows that $[\alpha * \beta]$ generates it. Since $\alpha * \beta$ is a loop in C, $j_* : \pi_1(C, x) \cong \mathbb{Z} \to \pi_1(X, x) \cong \mathbb{Z}$ is surjective, hence an isomorphism. ■

Theorem 10.18 (65.2)

If C is a simple closed curve in S^2 and p, q lie in different components of $S^2 - C$, then $j : C \to S^2 - p - q$ induces an isomorphism of fundamental groups.

Proof.

Step 1. if A is an arc from a to b and B an arc from b to c (a, b, c distinct), then $A \cup B$ contains an arc from a to c: with homeomorphisms $f : I \to A$, $g : I \to B$ from a to b and b to c, let t_0 be the least t with $f(t) \in B$ and $g(t_1) = f(t_0)$; then $f([0, t_0]) \cup g([t_1, 1])$ is such an arc.

Step 2. any two points of a connected open $U \subset \mathbb{R}^2$ are the end points of an arc in U: the relation "equal, or joined by an arc in U" is an equivalence relation by Step 1, with open classes (segments in a disc), hence one class.

Step 3. we build a complete graph on four vertices containing C as $a_1a_2a_3a_4a_1$. Work in $\mathbb{R}^2$ with 0 in the bounded component of $\mathbb{R}^2 - C$. Let a_1 be the largest point of C on the negative x-axis and a_3 the smallest on the positive x-axis; the open segment between them misses C and contains 0, so lies in the bounded component. Write $C = C_1 \cup C_2$, arcs from a_1 to a_3. Take a in the unbounded component. As C_1, C_2 do not separate, Step 2 gives injective paths α in $\mathbb{R}^2 - C_1$ and β in $\mathbb{R}^2 - C_2$ from a to 0. Let t_0 be least with $\alpha(t_0) \in C$ and $a_2 = \alpha(t_0)$, an interior point of C_2; similarly

$a_4 = \beta(t_1)$ interior to C_1. The arcs $\alpha([0,t_0])$ and $\beta([0,t_1])$ meet C only at their end points and otherwise lie in the unbounded component; by Step 1 their union contains an arc from a_2 to a_4 meeting $C \cup a_1a_3$ only in a_2, a_4. This arc, the segment a_1a_3 and C form the required graph. By Lemma 10.17 the theorem holds for some pair p, q in different components.

Step 4. it remains to move p (and, symmetrically, q) within its component. In $\mathbb{R}^2$-coordinates with $q = \infty$: let D be a simple closed curve, $0, p$ in the same component of $\mathbb{R}^2 - D$, and suppose $j : D \to \mathbb{R}^2 - 0$ induces an isomorphism. With $f(x) = x - p$, a homeomorphism $\mathbb{R}^2 - p \to \mathbb{R}^2 - 0$, and γ a path in $\mathbb{R}^2 - D$ from 0 to p, $F(x,t) = x - \gamma(t)$ is a homotopy in $\mathbb{R}^2 - 0$ from j to $f \circ k$, $k : D \to \mathbb{R}^2 - p$ the inclusion. By Corollary 9.35, $(f \circ k)_*$ and hence k_* are isomorphisms. ■

10.6 The Cauchy integral formula

Motivation. Complex analysis uses the winding number implicitly. Here the topological facts proved above are used to justify the Cauchy integral formula for simple closed curves.

For a loop f in $\mathbb{R}^2$ and $a \notin f(I)$, let $g(s) = (f(s) - a)/\|f(s) - a\|$, a loop in S^1, and $\tilde{g}$ a lifting to $\mathbb{R}$. The integer $n(f,a) = \tilde{g}(1) - \tilde{g}(0)$ is the **winding number** of f about a; it does not depend on the lifting, since any two differ by an integer constant. A **free homotopy** of loops is a map $F : I \times I \to X$ with $F(0,t) = F(1,t)$ for all t.

Lemma 10.19 (66.1)

Let f be a loop in $\mathbb{R}^2 - a$.
(a) $n(\bar{f}, a) = -n(f,a)$.
(b) If f is freely homotopic to f' through loops in $\mathbb{R}^2 - a$, then $n(f,a) = n(f',a)$.
(c) If a, b lie in the same component of $\mathbb{R}^2 - f(I)$, then $n(f,a) = n(f,b)$.

Proof. (a) Replacing s by $1 - s$ reverses the sign of $\tilde{g}(1) - \tilde{g}(0)$.

(b) For a free homotopy F, lift $G(s,t) = (F(s,t) - a)/\|F(s,t) - a\|$ to $\tilde{G} : I \times I \to \mathbb{R}$ (Lemma 9.13); $\tilde{G}(1,t) - \tilde{G}(0,t)$ is a continuous integer-valued function of t, hence constant.

(c) $n(f,a) = n(f - a, 0)$, and for a path γ in $\mathbb{R}^2 - f(I)$ from a to b, $f(s) - \gamma(t)$ is a free homotopy in $\mathbb{R}^2 - 0$ from $f - a$ to $f - b$. ■

A loop f is **simple** if $f(s) = f(s')$ only when $s = s'$ or $\{s, s'\} = \{0, 1\}$; its image is a simple closed curve.

Theorem 10.20 (66.2)

Let f be a simple loop in $\mathbb{R}^2$. If a is in the unbounded component of $\mathbb{R}^2 - f(I)$, $n(f,a) = 0$; if a is in the bounded component, $n(f,a) = \pm 1$.

Proof. We may take $a = 0$, and by rotating (a free homotopy) assume the base point x_0 lies on the positive x-axis. Let $C = f(I)$ and $h : S^1 \to C$ the homeomorphism with $h \circ p_0 = f$; $h_*[p_0]$ generates $\pi_1(C, x_0)$. If 0 is in the bounded component, $[f] = j_*h_*[p_0]$ generates $\pi_1(\mathbb{R}^2 - 0, x_0)$ by Theorem 10.18 (with ∞ in the other component). If 0 is in the unbounded component, $j \circ h$ is nulhomotopic (Lemma 10.2) and $[f]$ is trivial. The retraction $x \mapsto x/\|x\|$ induces an isomorphism onto $\pi_1(S^1, b_0)$, so $g = f/\|f\|$ represents a generator or the identity, i.e. (proof of Theorem 9.16) its lifting from 0 ends at ± 1 or at 0. ■

A simple loop is **counterclockwise** if $n(f,a) = +1$ for a in the bounded component and **clockwise** if it is -1.

Lemma 10.21 (66.3)

If f is a piecewise-differentiable loop in $\mathbb{C}$ and $a \notin f(I)$, then $n(f,a) = \dfrac{1}{2\pi i}\displaystyle\int_f \frac{dz}{z-a}$.

Proof. Let $r(s) = |f(s)-a|$, $g = (f-a)/r$, $\tilde{g}$ a lifting and $\theta = 2\pi\tilde{g}$, so $f(s)-a = r(s)e^{i\theta(s)}$ with r, θ piecewise differentiable (θ is locally a branch of the argument). Then

$$\int_f \frac{dz}{z-a} = \int_0^1 \frac{f'(s)}{f(s)-a}\,ds = \int_0^1 \Big(\frac{r'(s)}{r(s)} + i\theta'(s)\Big)ds = [\log r]_0^1 + i[\theta]_0^1 = 2\pi i\, n(f,a).$$

■

Theorem 10.22 (66.4, Cauchy integral formula, classical version)

Let C be a simple closed piecewise-differentiable curve in $\mathbb{C}$, B the bounded component of $\mathbb{R}^2 - C$, and F analytic on an open set $\Omega \supset B \cup C$. For $a \in B$,

$$F(a) = \pm\frac{1}{2\pi i}\int_C \frac{F(z)}{z-a}\,dz,$$

with $+$ if C is oriented counterclockwise and $-$ otherwise.

Proof. The general version (Ahlfors): if F is analytic on a region Ω, f a piecewise-differentiable loop in Ω with $n(f,b) = 0$ for every $b \notin \Omega$, and $a \in \Omega - f(I)$, then $n(f,a)F(a) = \frac{1}{2\pi i}\int_f \frac{F(z)}{z-a}dz$. For a parametrization f of C, every $b \notin \Omega$ is in the unbounded component, so $n(f,b) = 0$, and $n(f,a) = \pm 1$ for $a \in B$ by Theorem 10.20. ■

Chapter 11. The Seifert–van Kampen Theorem

11.1 Direct sums of abelian groups

Motivation. To compute fundamental groups of spaces glued from simpler pieces we need ways to combine groups. This section recalls direct sums for abelian groups, characterized by their universal property, as preparation for the noncommutative analogue.

Abelian groups are written additively. Subgroups G_α ($\alpha \in J$) of G **generate** G if every $x \in G$ is a finite sum $\sum x_\alpha$ with $x_\alpha \in G_\alpha$ (almost all 0); G is their **direct sum**, $G = \bigoplus G_\alpha$, if moreover this expression is unique. G is **free abelian** with **basis** $\{a_\alpha\}$ if each a_α generates an infinite cyclic subgroup G_α and $G = \bigoplus G_\alpha$.

Lemma 11.1 (67.1)

If $G = \bigoplus G_\alpha$, then G has the **extension property**: for every abelian group H and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique homomorphism $h : G \to H$ restricting to each h_α. Conversely, if the G_α generate G and the extension property holds, then $G = \bigoplus G_\alpha$.

Proof. If $G = \bigoplus G_\alpha$, define $h(\sum x_\alpha) = \sum h_\alpha(x_\alpha)$, well defined by uniqueness and clearly a homomorphism; it is forced, so unique. Conversely suppose $\sum x_\alpha = \sum y_\alpha$. For a fixed β, let $H = G_\beta$, $h_\beta = \mathrm{id}$ and $h_\alpha = 0$ for $\alpha \neq \beta$; applying the extension h to both sides gives $x_\beta = y_\beta$. ∎

Corollary 11.2 (67.2)

If $G = G_1 \oplus G_2$, $G_1 = \bigoplus_{\alpha \in J} H_\alpha$ and $G_2 = \bigoplus_{\beta \in K} H_\beta$ with $J \cap K = \emptyset$, then $G = \bigoplus_{\gamma \in J \cup K} H_\gamma$.

Proof. The H_γ generate G. Homomorphisms $h_\gamma : H_\gamma \to H$ extend to G_1 and to G_2, and these extend to G; uniqueness follows because the H_γ generate. Apply Lemma 11.1. ∎

Corollary 11.3 (67.3)

If $G = G_1 \oplus G_2$, then $G/G_2 \cong G_1$.

Proof. Extend $\mathrm{id} : G_1 \to G_1$ and $0 : G_2 \to G_1$ to $h : G \to G_1$; it is surjective with kernel G_2. ∎

Theorem 11.4 (67.4)

For any family $\{G_\alpha\}$ of abelian groups there are an abelian group G and monomorphisms $i_\alpha : G_\alpha \to G$ with $G = \bigoplus i_\alpha(G_\alpha)$ (the **external direct sum**).

Proof. Let G be the subgroup of $\prod G_\alpha$ of tuples with only finitely many nonzero entries, and $i_\beta(x)$ the tuple with x in coordinate β and 0 elsewhere. Each element is uniquely a finite sum of such tuples. ∎

Lemma 11.5 (67.5)

If $i_\alpha : G_\alpha \to G$ are monomorphisms and $G = \bigoplus i_\alpha(G_\alpha)$, then for any abelian H and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique $h : G \to H$ with $h \circ i_\alpha = h_\alpha$.

Proof. Apply Lemma 11.1 to the homomorphisms $h_\alpha \circ i_\alpha^{-1} : i_\alpha(G_\alpha) \to H$. ∎

Theorem 11.6 (67.6, uniqueness of direct sums)

If $G = \bigoplus i_\alpha(G_\alpha)$ and $G' = \bigoplus i'_\alpha(G_\alpha)$ with monomorphisms i_α, i'_α, there is a unique isomorphism $\phi : G \to G'$ with $\phi \circ i_\alpha = i'_\alpha$.

Proof. Lemma 11.5 gives $\phi : G \to G'$ with $\phi i_\alpha = i'_\alpha$ and $\psi : G' \to G$ with $\psi i'_\alpha = i_\alpha$. Then $\psi\phi i_\alpha = i_\alpha$, and by the uniqueness in Lemma 11.5 (with $H = G$, $h_\alpha = i_\alpha$), $\psi\phi = \mathrm{id}$; similarly $\phi\psi = \mathrm{id}$. ■

Lemma 11.7 (67.7)

Let $\{a_\alpha\}$ generate the abelian group G. Then G is free abelian with basis $\{a_\alpha\}$ iff for every abelian H and elements $y_\alpha \in H$ there is a homomorphism $h : G \to H$ with $h(a_\alpha) = y_\alpha$; such h is then unique.

Proof. Let G_α be the subgroup generated by a_α. If the condition holds, each a_α has infinite order: with $H = \mathbb{Z}$ and all $y_\alpha = 1$, $na_\alpha = 0$ forces $n = 0$. Homomorphisms $G_\alpha \to H$ correspond to choices of y_α, so the extension property of Lemma 11.1 holds, and $G = \bigoplus G_\alpha$. Conversely, if G is free abelian, $na_\alpha \mapsto ny_\alpha$ defines h_α on the infinite cyclic G_α; extend by Lemma 11.1. Uniqueness holds because the a_α generate. ■

Theorem 11.8 (67.8)

If G is free abelian with basis $\{a_1, \dots, a_n\}$, then n is determined by G.

Proof. $G \cong \mathbb{Z}^n$, and $2G = \{2x\}$ consists of the elements $\sum n_i a_i$ with all n_i even. The map $\sum n_i a_i \mapsto (n_i \bmod 2)$ is a surjection $G \to (\mathbb{Z}/2)^n$ with kernel $2G$, so $G/2G$ has 2^n elements. The number $|G/2G|$ depends only on G. ■

n is the **rank** of G.

11.2 Free products of groups

Motivation. The fundamental group of a figure eight is not abelian: going around one loop and then the other differs from the reverse. The free product is the group-theoretic way of combining two groups with no relations between them, and it is what van Kampen's theorem produces.

Let G be a group and $\{G_\alpha\}$ subgroups generating G with $G_\alpha \cap G_\beta = \{1\}$ for $\alpha \neq \beta$. A **word** is a finite sequence $(g_1, \dots, g_n)$ of elements of $\bigcup G_\alpha$, representing the product $g_1 \cdots g_n$; it is **reduced** if no $g_i = 1$ and no two consecutive g_i, g_{i+1} lie in the same G_α. Every element of G is represented by a reduced word (combine adjacent letters in one group and delete 1's), the identity by the empty word. G is the **free product** of the G_α, $G = \prod^* G_\alpha$, if each element is represented by only one reduced word. It suffices that no nonempty reduced word represent 1: if reduced words $(g_1, \dots, g_n)$ and $(h_1, \dots, h_m)$ represent the same element, then $g_1 \cdots g_n h_m^{-1} \cdots h_1^{-1} = 1$; if g_n, h_m lie in different groups this word is reduced and nonempty, if they lie in one group and differ, merging them gives a nonempty reduced word, and if $g_n = h_m$ we cancel and induct.

Lemma 11.9 (68.1)

If $G = \prod^* G_\alpha$, then for every group H and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique homomorphism $h : G \to H$ restricting to each h_α.

Proof. For x represented by the reduced word $(g_1, \dots, g_n)$ with $g_i \in G_{\alpha_i}$, set $h(x) = h_{\alpha_1}(g_1) \cdots h_{\alpha_n}(g_n)$; this is well defined by uniqueness. The same formula, applied to any word, gives the same value, since combining adjacent letters of one group or deleting 1's does not change the product of the images. The concatenation of words representing x and y represents xy, so $h(xy) = h(x)h(y)$. Uniqueness holds because the G_α generate G. ∎

Theorem 11.10 (68.2)

For any family $\{G_\alpha\}$ of groups there are a group G and monomorphisms $i_\alpha : G_\alpha \to G$ such that G is the free product of the subgroups $i_\alpha(G_\alpha)$ (the **external free product**).

Proof. We may assume the sets $G_\alpha - \{1\}$ disjoint. Let W be the set of reduced words in their union (including the empty word). For $g \in G_\alpha$ define $\pi_g : W \to W$: $\pi_1 = \mathrm{id}$; for $g \neq 1$ and $w = (g_1, \dots, g_n)$, $\pi_g(w) = (g, g_1, \dots, g_n)$ if w is empty or $g_1 \notin G_\alpha$; $(gg_1, g_2, \dots, g_n)$ if $g_1 \in G_\alpha$ and $gg_1 \neq 1$; and $(g_2, \dots, g_n)$ if $gg_1 = 1$. Checking the cases shows $\pi_g \pi_{g'} = \pi_{gg'}$ for $g, g' \in G_\alpha$ (both sides only modify the first letter when it lies in G_α). Hence each π_g is a bijection with inverse $\pi_{g^{-1}}$, and $i_\alpha(g) = \pi_g$ is a homomorphism into the group of permutations of W; it is injective because $\pi_g(\emptyset) = (g)$. Let G be the group generated by all π_g. If $(g_1, \dots, g_n)$ is a nonempty reduced word, then $\pi_{g_1} \cdots \pi_{g_n}(\emptyset) = (g_1, \dots, g_n) \neq \emptyset$, so it does not represent the identity. Also $i_\alpha(G_\alpha) \cap i_\beta(G_\beta) = \{1\}$ for the same reason. ∎

Lemma 11.11 (68.3)

If $i_\alpha : G_\alpha \to G$ are monomorphisms and G is the free product of the $i_\alpha(G_\alpha)$, then for every group H and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique $h : G \to H$ with $h \circ i_\alpha = h_\alpha$.

Proof. Apply Lemma 11.9 to $h_\alpha \circ i_\alpha^{-1}$. ∎

Theorem 11.12 (68.4, uniqueness of free products)

Let $i_\alpha : G_\alpha \to G$ and $i'_\alpha : G_\alpha \to G'$ be monomorphisms whose images generate G and G', both with the extension property of Lemma 11.11. Then there is a unique isomorphism $\phi : G \to G'$ with $\phi \circ i_\alpha = i'_\alpha$.

Proof. As in Theorem 11.6: the extensions ϕ, ψ satisfy $\psi\phi i_\alpha = i_\alpha$, so $\psi\phi = \mathrm{id}$ by uniqueness, and likewise $\phi\psi = \mathrm{id}$. ∎

Lemma 11.13 (68.5)

If homomorphisms $i_\alpha : G_\alpha \to G$ satisfy the extension condition of Lemma 11.11, then each i_α is a monomorphism and G is the free product of the $i_\alpha(G_\alpha)$.

Proof. For a fixed β, take $H = G_\beta$, $h_\beta = \mathrm{id}$ and h_α trivial for $\alpha \neq \beta$; the extension h has $h \circ i_\beta = \mathrm{id}$, so i_β is injective. Let G' with i'_α be the external free product (Theorem 11.10), which has the extension property. Extensions give $\phi : G \to G'$ with $\phi i_\alpha = i'_\alpha$ and $\psi : G' \to G$ with $\psi i'_\alpha = i_\alpha$. By the uniqueness in the extension condition for G, $\psi\phi = \mathrm{id}_G$, and since G' is generated by the $i'_\alpha(G_\alpha)$, $\phi\psi = \mathrm{id}$. So ϕ is an isomorphism carrying $i_\alpha(G_\alpha)$ onto $i'_\alpha(G_\alpha)$, and G is the free product of the $i_\alpha(G_\alpha)$. ∎

Corollary 11.14 (68.6)

If $G = G_1 * G_2$ with $G_1 = \prod^*_{\alpha \in J} H_\alpha$, $G_2 = \prod^*_{\beta \in K} H_\beta$ and $J \cap K = \emptyset$, then $G = \prod^*_{\gamma \in J \cup K} H_\gamma$.

Proof. Homomorphisms $h_\gamma : H_\gamma \to H$ extend to G_1 and G_2, and then to G; the extension is unique since the H_γ generate G. Apply Lemma 11.13 to the inclusions. ■

Theorem 11.15 (68.7)

Let $G = G_1 * G_2$, $N_i \lhd G_i$, and N the least normal subgroup of G containing N_1 and N_2. Then $G/N \cong (G_1/N_1) * (G_2/N_2)$.

Proof. Let $F = (G_1/N_1) * (G_2/N_2)$. The projections $G_i \to G_i/N_i \to F$ extend to $\phi : G \to F$, which is surjective and kills N_1, N_2, hence N; it induces $\bar{\phi} : G/N \to F$. The maps $G_i/N_i \to G/N$, $gN_i \mapsto gN$, are well-defined homomorphisms ($N_i \subset N$) and extend to $\psi : F \to G/N$. On generators gN ($g \in G_i$) and gN_i, $\psi\bar{\phi}$ and $\bar{\phi}\psi$ are the identity; hence they are the identity. ■

Corollary 11.16 (68.8)

If N is the least normal subgroup of $G_1 * G_2$ containing G_1, then $(G_1 * G_2)/N \cong G_2$.

Proof. Take $N_1 = G_1$ and $N_2 = \{1\}$ in Theorem 11.15. ■

Lemma 11.17 (68.9)

The least normal subgroup N of G containing $S \subset G$ is generated by the conjugates of elements of S.

Proof. The subgroup N' generated by these conjugates contains S and is normal, since $g(x_1 \cdots x_k)g^{-1} = (gx_1g^{-1}) \cdots (gx_kg^{-1})$ and a conjugate of a conjugate is a conjugate. Every normal subgroup containing S contains N', so $N = N'$. ■

11.3 Free groups

Motivation. Free groups are free products of copies of $\mathbb{Z}$. They are the fundamental groups of graphs, and every group is a quotient of a free group, which is how presentations of fundamental groups arise.

G is a **free group** with **system of free generators** $\{a_\alpha\}$ if each a_α generates an infinite cyclic subgroup G_α and $G = \prod^* G_\alpha$.

Lemma 11.18 (69.1)

If G is free on $\{a_\alpha\}$, then for every group H and elements $y_\alpha \in H$ there is a unique homomorphism $h : G \to H$ with $h(a_\alpha) = y_\alpha$. Conversely, if the a_α generate G and this extension condition holds, G is free on $\{a_\alpha\}$.

Proof. If G is free, $a_\alpha^n \mapsto y_\alpha^n$ defines homomorphisms on the infinite cyclic G_α; extend by Lemma 11.9. Conversely, with $H = \mathbb{Z}$ and $y_\alpha = 1$, $a_\alpha^n = 1$ forces $n = 0$, so G_α is infinite cyclic; homomorphisms $G_\alpha \to H$ correspond to the choice of y_α, so the extension condition of Lemma 11.11 holds for the inclusions, and Lemma 11.13 applies. ■

Theorem 11.19 (69.2)

If $G = G_1 * G_2$ with G_1, G_2 free on $\{a_\alpha\}_{\alpha\in J}$ and $\{a_\alpha\}_{\alpha\in K}$, $J \cap K = \emptyset$, then G is free on $\{a_\alpha\}_{\alpha\in J\cup K}$.

Proof. Corollary 11.14. ■

If G is free on $\{a_\alpha\}$ and $\{r_\beta\} \subset G$, and N is the least normal subgroup containing the r_β, a group isomorphic to G/N has the **presentation** with generators a_α and relations r_β. The **commutator subgroup** $[G, G]$ is generated by all commutators $[x, y] = xyx^{-1}y^{-1}$.

Lemma 11.20 (69.3)

$[G, G]$ is normal and $G/[G, G]$ is abelian. Every homomorphism $h : G \to H$ into an abelian group has $[G, G] \subset \ker h$, so it induces $k : G/[G, G] \to H$.

Proof. $g[x, y]g^{-1} = [gxg^{-1}, gyg^{-1}]$, so conjugates of products of commutators are products of commutators. $xy(yx)^{-1} = [x, y]$, so cosets of x and y commute. If H is abelian, $h([x, y]) = 1$. ■

Theorem 11.21 (69.4)

If G is free on $\{a_\alpha\}$, then $G/[G, G]$ is free abelian with basis the cosets $[a_\alpha]$.

Proof. The cosets generate. Given y_α in an abelian H, Lemma 11.18 gives h with $h(a_\alpha) = y_\alpha$, which induces k with $k([a_\alpha]) = y_\alpha$ (Lemma 11.20). Apply Lemma 11.7. ■

Corollary 11.22 (69.5)

If G is free on n generators, every system of free generators of G has n elements.

Proof. $G/[G, G]$ is free abelian of rank equal to the number of free generators (Theorem 11.21), and the rank is unique (Theorem 11.8). (An infinite system cannot occur: $G/[G, G]$ would then not be finitely generated.) ■

11.4 The Seifert–van Kampen theorem

Motivation. This is the main computational tool for fundamental groups: if $X = U \cup V$ with $U, V, U \cap V$ path connected, then $\pi_1(X)$ is the free product of $\pi_1(U)$ and $\pi_1(V)$ with the images of $\pi_1(U \cap V)$ identified.

Theorem 11.23 (70.1, Seifert–van Kampen theorem)

Let $X = U \cup V$ with U, V open and U, V, $U \cap V$ path connected, $x_0 \in U \cap V$. Let i_1, i_2 be induced by the inclusions of $U \cap V$ into U and V, and j_1, j_2 by the inclusions of U and V into X. If $\phi_1 : \pi_1(U, x_0) \to H$ and $\phi_2 : \pi_1(V, x_0) \to H$ are homomorphisms with $\phi_1 \circ i_1 = \phi_2 \circ i_2$, there is a unique homomorphism $\Phi : \pi_1(X, x_0) \to H$ with $\Phi \circ j_1 = \phi_1$ and $\Phi \circ j_2 = \phi_2$.

Proof. Uniqueness: $\pi_1(X, x_0)$ is generated by the images of j_1, j_2 (Theorem 9.38). Write $[f]_U, [f]_V$ for path classes in U, V and $[f]$ for classes in X.

Step 1. for a loop f at x_0 in U or V, let $\rho(f) = \phi_1([f]_U)$ or $\phi_2([f]_V)$; if f lies in $U \cap V$ the two agree since $\phi_1 i_1 = \phi_2 i_2$. Then (1) $[f]_U = [g]_U$ or $[f]_V = [g]_V$ implies $\rho(f) = \rho(g)$;

(2) if f, g both lie in U or both in V, $\rho(f * g) = \rho(f)\rho(g)$.

Step 2. for each $x \in X$ choose a path α_x from x_0 to x: constant if $x = x_0$, in $U \cap V$ if $x \in U \cap V$, and in U (resp. V) if x lies only in U (resp. V). For a path f in U or V from x to y, $L(f) = \alpha_x * f * \bar{\alpha}_y$ is a loop at x_0 in U or V; let $\sigma(f) = \rho(L(f))$. Then σ extends ρ (for a loop at x_0, $L(f) \simeq_p f$ in U or V), satisfies (1), and satisfies (2) whenever $f * g$ is defined, since $L(f) * L(g) \simeq_p L(f * g)$ in U or V.

Step 3. for an arbitrary path f in X, choose $0 = s_0 < \cdots < s_n = 1$ with each $[s_{i-1}, s_i]$ mapped into U or V, let f_i be the reparametrized pieces, and set $\tau(f) = \sigma(f_1) \cdots \sigma(f_n)$. Adding a point p to the subdivision replaces $\sigma(f_i)$ by $\sigma(f_i')\sigma(f_i'')$, which is equal since $f_i \simeq_p f_i' * f_i''$ in U or V; so τ is well defined (any two subdivisions have a common refinement) and extends σ.

Step 4. if $f \simeq_p g$ in X, then $\tau(f) = \tau(g)$. Special case: a path homotopy F for which some subdivision $s_0 < \cdots < s_n$ has each $R_i = [s_{i-1}, s_i] \times I$ mapped into U or V. Let $\beta_i(t) = F(s_i, t)$, so β_0, β_n are constant. In R_i the path along the bottom and right edges and the path along the left and top edges are path homotopic (convexity); applying F gives $f_i * \beta_i \simeq_p \beta_{i-1} * g_i$ in U or V, so $\sigma(f_i)\sigma(\beta_i) = \sigma(\beta_{i-1})\sigma(g_i)$. Also $\sigma(\beta_0) = \sigma(\beta_n) = 1$ (from $\beta * \beta = \beta$ up to path homotopy for a constant path). Multiplying, $\tau(f) = \sigma(f_1) \cdots \sigma(f_n) = \sigma(g_1) \cdots \sigma(g_n) = \tau(g)$. General case: subdivide $I \times I$ into rectangles $[s_{i-1}, s_i] \times [t_{j-1}, t_j]$ mapped into U or V and apply the special case to consecutive paths $f_j(s) = F(s, t_j)$.

Step 5. $\tau(f * g) = \tau(f)\tau(g)$: use a subdivision of I containing $\frac{1}{2}$ for $f * g$; its two halves give subdivisions for f and g, and the factors match.

Step 6. define $\Phi([f]) = \tau(f)$. By Steps 4 and 5 it is a well-defined homomorphism, and for a loop f in U, $\Phi(j_1[f]_U) = \tau(f) = \rho(f) = \phi_1([f]_U)$; similarly for V. ■

Theorem 11.24 (70.2, Seifert–van Kampen theorem, classical version)

Under the same hypotheses, let $j : \pi_1(U, x_0) * \pi_1(V, x_0) \to \pi_1(X, x_0)$ extend j_1 and j_2. Then j is surjective, and its kernel is the least normal subgroup N containing all elements $i_1(g)^{-1} i_2(g)$, $g \in \pi_1(U \cap V, x_0)$.

Proof. Surjectivity is Theorem 9.38. Since $j_1 i_1 = j_2 i_2$ (both are induced by the inclusion of $U \cap V$ into X), each $i_1(g)^{-1} i_2(g)$ lies in $\ker j$, so $N \subset \ker j$ and j induces an epimorphism $k : F/N \to \pi_1(X, x_0)$, F the free product. Let $H = F/N$ and ϕ_1, ϕ_2 the inclusions into F followed by projection. Then $\phi_1 i_1(g) = i_1(g)N = i_2(g)N = \phi_2 i_2(g)$, so Theorem 11.23 gives Φ with $\Phi j_1 = \phi_1$, $\Phi j_2 = \phi_2$. For $g \in \pi_1(U, x_0)$, $\Phi k(gN) = \Phi j_1(g) = gN$, and similarly for V; these cosets generate H, so $\Phi \circ k = \mathrm{id}$ and k is injective. Hence $\ker j = N$. ■

Corollary 11.25 (70.3)

If moreover $U \cap V$ is simply connected, then $\pi_1(U, x_0) * \pi_1(V, x_0) \cong \pi_1(X, x_0)$.

Corollary 11.26 (70.4)

If V is simply connected, then $\pi_1(U, x_0)/N \cong \pi_1(X, x_0)$, where N is the least normal subgroup of $\pi_1(U, x_0)$ containing the image of i_1.

Proof. In Theorem 11.24, N is trivial in the first case; in the second, the free product is $\pi_1(U, x_0)$ and $i_1(g)^{-1} i_2(g) = i_1(g)^{-1}$. ■

Example 11.1

A theta space $X = A \cup B \cup C$ (arcs meeting in their end points) has free fundamental group of rank 2: with a, b interior points of A, B, let $U = X - a$, $V = X - b$. $U \cap V$ is contractible, while U and V deformation retract onto the circles $B \cup C$ and $A \cup C$. Apply Corollary 11.25.

11.5 The fundamental group of a wedge of circles

Motivation. The first application: a wedge of n circles has fundamental group free on n generators.

A Hausdorff space X that is the union of subspaces $S_1, \ldots, S_n$, each homeomorphic to S^1, with $S_i \cap S_j = \{p\}$ for $i \neq j$, is the **wedge** of these circles.

Theorem 11.27 (71.1)

If X is the wedge of $S_1, \ldots, S_n$ and f_i is a loop in S_i representing a generator of $\pi_1(S_i, p)$, then $\pi_1(X, p)$ is free with free generators $[f_1], \ldots, [f_n]$.

Proof. Induction on n, the case $n = 1$ being clear. Choose $q_i \in S_i - p$ and let $W_i = S_i - q_i$, $U = S_1 \cup W_2 \cup \cdots \cup W_n$, $V = W_1 \cup S_2 \cup \cdots \cup S_n$. Then $U \cap V = W_1 \cup \cdots \cup W_n$; U, V, $U \cap V$ are open and path connected. Each W_i is an open arc deformation retracting onto p; these deformations fit together, by the pasting lemma on the closed sets $W_i \times I$ of $(U \cap V) \times I$, to a deformation retraction of $U \cap V$ onto p. So $U \cap V$ is simply connected and $\pi_1(X, p) \cong \pi_1(U, p) * \pi_1(V, p)$ by Corollary 11.25. Similarly U deformation retracts onto S_1 and V onto $S_2 \cup \cdots \cup S_n$; so $\pi_1(U, p)$ is infinite cyclic generated by $[f_1]$ and, by induction, $\pi_1(V, p)$ is free on $[f_2], \ldots, [f_n]$. Apply Theorem 11.19. ■

A space X that is the union of subspaces X_α has the topology **coherent** with them if $C \subset X$ is closed whenever each $C \cap X_\alpha$ is closed in X_α. If X is the union of subspaces S_α ($\alpha \in J$) homeomorphic to S^1 with $S_\alpha \cap S_\beta = \{p\}$ for $\alpha \neq \beta$, and the topology is coherent with them, X is the **wedge** of the circles S_α (for finite J this agrees with the earlier definition, a finite union of closed sets being coherent).

Lemma 11.28 (71.2)

A wedge X of circles S_α is normal, and every compact subspace lies in the union of finitely many S_α.

Proof. Points are closed by coherence. Let A, B be disjoint closed with $p \notin B$. In each S_α choose disjoint open sets $U_\alpha \supset \{p\} \cup (A \cap S_\alpha)$ and $V_\alpha \supset B \cap S_\alpha$ (S_α is normal). $U = \bigcup U_\alpha$ and $V = \bigcup V_\alpha$ are disjoint, and $U \cap S_\alpha = U_\alpha$ (all U_β contain p, the only common point), $V \cap S_\alpha = V_\alpha$ (no V_β contains p); by coherence U, V are open. If C is compact, choose $x_\alpha \in C \cap (S_\alpha - p)$ whenever possible. Every subset of $D = \{x_\alpha\}$ meets each S_α in at most one point, hence is closed; so D is closed and discrete in C, and as C is limit point compact, D is finite. ■

Theorem 11.29 (71.3)

If X is the wedge of circles S_α and f_α represents a generator of $\pi_1(S_\alpha, p)$, then $\pi_1(X, p)$ is free with free generators $\{[f_\alpha]\}$.

Proof. Let G_α be the image of $\pi_1(S_\alpha, p)$. A loop or a path homotopy in X has compact image, so lies in a finite union of circles (Lemma 11.28). Hence: the G_α generate $\pi_1(X, p)$ (by Theorem 11.27 in a finite wedge); each $\pi_1(S_\alpha, p) \to G_\alpha$ is injective, so G_α is infinite cyclic; and a nonempty reduced

word in the G_α representing the identity would give a loop nulhomotopic in some finite wedge, contradicting Theorem 11.27. ■

Example 11.2

The **infinite earring**, the union $X \subset \mathbb{R}^2$ of the circles C_n of radius $1/n$ centred at $(1/n, 0)$, is not a wedge: its intersection with the positive x-axis meets each C_n in one point but is not closed. Indeed the classes $[f_n]$ do not generate $\pi_1(X, p)$: let g run through f_n on $[1/(n+1), 1/n]$, with $g(0) = p$, a continuous loop. For $N > n$, the retraction of X onto C_N (collapsing the other circles to p) kills $[f_1], \ldots, [f_n]$ but sends $[g]$ to the generator $[f_N]$; so $[g]$ is not in the subgroup generated by $[f_1], \ldots, [f_n]$ for any n.

Lemma 11.30 (71.4)

For every index set J there is a wedge of circles S_α, $\alpha \in J$.

Proof. Give J the discrete topology, let $E = S^1 \times J$, and let X be the quotient of E collapsing $P = b_0 \times J$ to a point p, with quotient map π; put $S_\alpha = \pi(S^1 \times \alpha)$. If C is closed in $S^1 \times \alpha$, then $\pi^{-1}\pi(C)$ is C or $C \cup P$, closed, so $\pi(C)$ is closed. Hence S_α is closed and π maps $S^1 \times \alpha$ homeomorphically onto S_α. If $D \cap S_\alpha$ is closed for each α, then $\pi^{-1}(D) \cap (S^1 \times \alpha)$ is closed for each α, so $\pi^{-1}(D)$ is closed and D is closed: the topology is coherent. ■

11.6 Adjoining a two-cell

Motivation. Attaching a disc along a loop kills that loop. In group terms, adjoining a 2-cell adds one relation, so cell structures translate directly into presentations of π_1.

Theorem 11.31 (72.1)

Let X be Hausdorff, $A \subset X$ closed and path connected, and $h : B^2 \to X$ continuous, mapping $\operatorname{Int} B^2$ bijectively onto $X - A$ and S^1 into A. Let $p \in S^1$, $a = h(p)$ and $k = h|S^1 : (S^1, p) \to (A, a)$. Then $i_* : \pi_1(A, a) \to \pi_1(X, a)$ induced by inclusion is surjective, and its kernel is the least normal subgroup containing $k_*(\pi_1(S^1, p))$.

Proof.

Step 1. with $x_0 = h(0)$ and $U = X - x_0$, A is a deformation retract of U. Let $C = h(B^2)$, closed in X, and $\pi : B^2 \to C$ the restriction of h; $\pi \times \mathrm{id} : B^2 \times I \to C \times I$ is closed (compact to Hausdorff), hence a quotient map, and so is its restriction to the saturated open set $(B^2 - 0) \times I$. The radial deformation retraction of $B^2 - 0$ onto S^1 (which fixes S^1) therefore induces a deformation retraction of $C - x_0$ onto $\pi(S^1)$; extend it by keeping A fixed. By the pasting lemma (the closed sets $C - x_0$ and A of U overlap in $\pi(S^1)$, which stays fixed) this is a deformation retraction of U onto A. So $\pi_1(A, a) \to \pi_1(U, a)$ is an isomorphism (Theorem 9.33), and it suffices to show: $\pi_1(U, a) \to \pi_1(X, a)$ is surjective with kernel the least normal subgroup containing $[g]$, $g = h \circ f$, f a generator of $\pi_1(S^1, p)$.

Step 2. let $V = X - A = h(\operatorname{Int} B^2)$, open. $\pi|\operatorname{Int} B^2$ is a quotient map (restriction to a saturated open set) and bijective, so a homeomorphism; V is simply connected. $U \cap V = V - x_0 \cong \operatorname{Int} B^2 - 0$ is path connected with infinite cyclic fundamental group, and U is path connected. For $b \in U \cap V$, Corollary 11.26 shows $\pi_1(U, b) \to \pi_1(X, b)$ is surjective with kernel the least normal subgroup containing the image of $\pi_1(U \cap V, b)$.

Step 3. let q be the midpoint of the segment from 0 to p, $b = h(q)$, f_0 a loop in $\operatorname{Int} B^2 - 0$ at q generating its fundamental group, $g_0 = h \circ f_0$, γ the segment from q to p and $\delta = h \circ \gamma$. The isomorphisms $\hat{\delta}$ commute with the homomorphisms induced by inclusion $U \to X$, so $\pi_1(U, a) \to \pi_1(X, a)$ is surjective with kernel the least normal subgroup containing $\hat{\delta}([g_0])$. Now $\bar{\gamma} * f_0 * \gamma$ generates $\pi_1(B^2 - 0, p)$, so it is path homotopic in $B^2 - 0$ to f or $\bar{f}$; applying h, $\hat{\delta}([g_0]) = [g]^{\pm 1}$ in $\pi_1(U, a)$. ■

A space homeomorphic to B^2 is a **2-cell**; X is obtained from A by **adjoining a 2-cell**.

11.7 The fundamental groups of the torus and the dunce cap

Motivation. Applying the two-cell result to standard polygon models gives presentations for the torus, $\langle a, b \mid aba^{-1}b^{-1}\rangle \cong \mathbb{Z}^2$, and for the dunce cap.

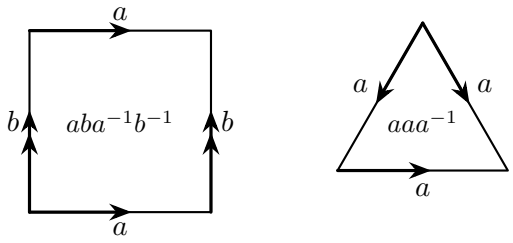


Figure. *Left: the torus is a square with boundary word $aba^{-1}b^{-1}$, so $\pi_1 = \langle a, b \mid aba^{-1}b^{-1}\rangle \cong \mathbb{Z} \times \mathbb{Z}$. Right: a dunce cap is a triangle with all three edges identified as a, a, a^{-1} (reading around the boundary), so $\pi_1 = \langle a \mid a^2a^{-1}\rangle$ is trivial.*

Theorem 11.32 (73.1)

The fundamental group of the torus has a presentation with two generators α, β and one relation $\alpha\beta\alpha^{-1}\beta^{-1}$.

Proof. Let $h : I^2 \to T = S^1 \times S^1$ be the restriction of $p \times p$, $A = h(\operatorname{Bd} I^2)$, a wedge of two circles, and $a = h(0, 0)$; the hypotheses of Theorem 11.31 hold ($I^2 \cong B^2$). With $a_0(t) = (t, 0)$, $b_0(t) = (0, t)$, $a_1(t) = (t, 1)$, $b_1(t) = (1, t)$, the loops $\alpha = h \circ a_0$, $\beta = h \circ b_0$ are free generators of $\pi_1(A, a)$ (Theorem 11.27). The loop $f = a_0 * (b_1 * (\bar{a}_1 * \bar{b}_0))$ generates $\pi_1(\operatorname{Bd} I^2, p)$, and $h \circ f = \alpha * (\beta * (\bar{\alpha} * \bar{\beta}))$. Apply Theorem 11.31. ■

Corollary 11.33 (73.2)

The fundamental group of the torus is free abelian of rank 2.

Proof. Let G be free on α, β and N the least normal subgroup containing $\alpha\beta\alpha^{-1}\beta^{-1}$. It is a commutator, so $N \subset [G, G]$; and G/N is abelian, being generated by the commuting cosets $\alpha N, \beta N$, so $[G, G] \subset N$. Hence $G/N = G/[G, G]$, free abelian of rank 2 (Theorem 11.21). ■

For $n > 1$ let r be rotation of S^1 by $2\pi/n$. The **n-fold dunce cap** X is the quotient of B^2 identifying each $x \in S^1$ with $r(x), \ldots, r^{n-1}(x)$. The quotient map π is closed: for closed C, $\pi^{-1}\pi(C) = C \cup r(C_0) \cup \cdots \cup r^{n-1}(C_0)$ with $C_0 = C \cap S^1$. So X is compact, and Hausdorff by the following lemma.

Lemma 11.34 (73.3)

If $\pi : E \to X$ is a closed quotient map and E is normal, so is X.

Proof. Points of X are closed since points of E are. For disjoint closed $A, B \subset X$, take disjoint open $U \supset \pi^{-1}(A)$, $V \supset \pi^{-1}(B)$. The sets $U_0 = X - \pi(E - U)$ and $V_0 = X - \pi(E - V)$ are open (as π is closed), contain A and B, and are disjoint: if $x \in U_0$, then $\pi^{-1}(x) \subset U$, and if $x \in V_0$, $\pi^{-1}(x) \subset V$. ■

The 2-fold dunce cap is homeomorphic to P^2: following the homeomorphism of B^2 onto the upper hemisphere by $S^2 \to P^2$ gives a closed continuous surjection that is injective on $\operatorname{Int} B^2$ and identifies x with $-x$ on S^1.

Theorem 11.35 (73.4)

The fundamental group of the n-fold dunce cap is cyclic of order n.

Proof. Let $h : B^2 \to X$ be the quotient map, $A = h(S^1)$, $p = (1, 0)$, $a = h(p)$. h maps the arc of S^1 from p to $r(p)$ onto A, identifying only its end points, so A is a simple closed curve and, with $\gamma(t) = (\cos 2\pi t/n, \sin 2\pi t/n)$, $\alpha = h \circ \gamma$ generates $\pi_1(A, a) \cong \mathbb{Z}$. The loop $f = \gamma * (r \circ \gamma) * \cdots * (r^{n-1} \circ \gamma)$ generates $\pi_1(S^1, p)$, and since $h \circ r^m = h$ on S^1, $h \circ f$ is the n-fold product of α. By Theorem 11.31, $\pi_1(X, a) \cong \mathbb{Z}/n\mathbb{Z}$. ■

Chapter 12. Classification of Surfaces

12.1 Fundamental groups of surfaces

Motivation. Every compact surface is obtained by gluing the edges of a polygon in pairs, and the gluing pattern gives a presentation of its fundamental group. This is the tool for telling surfaces apart.

A **polygonal region** P in the plane is the convex hull of points $p_0, \dots, p_{n-1}$ ($n \geq 3$) lying on a circle in counterclockwise order; its **edges** are the segments $p_{i-1}p_i$ ($p_n = p_0$). An **orientation** of an edge is a choice of initial point; a **labelling** assigns a label to each edge. If edges with the same label are pasted by the positive linear maps respecting orientations, we obtain a quotient space X. A labelling and orientation are recorded by a **labelling scheme** $w = (a_{i_1})^{\epsilon_1} \cdots (a_{i_n})^{\epsilon_n}$: the k-th edge has label a_{i_k} and is oriented counterclockwise if $\epsilon_k = +1$, clockwise if $\epsilon_k = -1$. Several disjoint polygonal regions with one scheme each are handled the same way. For example, $aba^{-1}b^{-1}$ gives the torus, $abab$ the projective plane (the 2-fold dunce cap), and $aa^{-1}bb^{-1}$ the sphere.

Theorem 12.1 (74.1)

The space X obtained from finitely many polygonal regions by pasting edges according to a labelling scheme is compact Hausdorff.

Proof. Compactness is clear. By Lemma 11.34 (the regions are normal) it suffices that the quotient map π is closed, i.e. $\pi^{-1}\pi(C)$ is closed for closed C. For an edge e, the set $\pi^{-1}\pi(C) \cap e$ is the union of $C \cap e$ and the images $h_i(C \cap e_i)$ under the pasting homeomorphisms $h_i : e_i \to e$ of the edges e_i pasted to e; this is compact, hence closed. And $\pi^{-1}\pi(C)$ is the union of C and these finitely many sets. ∎

Theorem 12.2 (74.2)

Let X be obtained from a polygonal region P by a scheme $w = (a_{i_1})^{\epsilon_1} \cdots (a_{i_n})^{\epsilon_n}$, with quotient map π. If π maps all vertices to one point x_0 and $a_1, \dots, a_k$ are the distinct labels, then $\pi_1(X, x_0)$ is the quotient of the free group on $\alpha_1, \dots, \alpha_k$ by the least normal subgroup containing $(\alpha_{i_1})^{\epsilon_1} \cdots (\alpha_{i_n})^{\epsilon_n}$.

Proof. $A = \pi(\operatorname{Bd} P)$ is a wedge of k circles, one for each label (each edge image is an arc whose end points are identified to x_0). For each i choose an edge labelled a_i, let f_i be the linear path along it in the counterclockwise direction and $g_i = \pi \circ f_i$; by Theorem 11.27, the $[g_i]$ (or their inverses) are free generators of $\pi_1(A, x_0)$. $P \cong B^2$, and π maps $\operatorname{Int} P$ bijectively onto $X - A$. The loop f going once counterclockwise around $\operatorname{Bd} P$ generates $\pi_1(\operatorname{Bd} P)$, and $\pi \circ f$ is the product $g_{i_1}^{\epsilon_1} * \cdots * g_{i_n}^{\epsilon_n}$, where g^{-1} means $\bar{g}$ (the edge labelled a_{i_j} with $\epsilon_j = -1$ is traversed against its orientation). Theorem 11.31 gives the result. ∎

The **n-fold torus** $T_n = T\# \cdots \# T$ is obtained from a $4n$-gon by the scheme $(a_1b_1a_1^{-1}b_1^{-1}) \cdots (a_nb_na_n^{-1}b_n^{-1})$; for $n = 2$ it is the double torus of Theorem 9.46. The **m-fold projective plane** P_m is obtained from a $2m$-gon by $(a_1a_1) \cdots (a_ma_m)$ ($m > 1$); set $P_1 = P^2$.

Theorem 12.3 (74.3)

$\pi_1(T_n, x_0)$ is the quotient of the free group on $\alpha_1, \beta_1, \dots, \alpha_n, \beta_n$ by the least normal subgroup containing $[\alpha_1, \beta_1] \cdots [\alpha_n, \beta_n]$, where $[\alpha, \beta] = \alpha\beta\alpha^{-1}\beta^{-1}$.

Proof. By Theorem 12.2 it suffices that all vertices are identified. In a block $aba^{-1}b^{-1}$ on vertices $v_0, \dots, v_4$: the edges labelled a paste v_0 to v_3 and v_1 to v_2; those labelled b paste v_1 to v_4 and v_2 to v_3. So $v_0, \dots, v_4$ are identified, and consecutive blocks share a vertex. ■

Theorem 12.4 (74.4)

$\pi_1(P_m, x_0)$ is the quotient of the free group on $\alpha_1, \dots, \alpha_m$ by the least normal subgroup containing $\alpha_1^2\alpha_2^2 \cdots \alpha_m^2$.

Proof. In a block aa on v_0, v_1, v_2, the two edges paste v_0 to v_1 and v_1 to v_2; consecutive blocks share a vertex. Apply Theorem 12.2. ■

12.2 Homology of surfaces

Motivation. Presentations are hard to compare directly. Abelianizing the fundamental group (first homology) gives a finitely generated abelian group, which is easy to compare and still distinguishes all the standard surfaces.

For path-connected X let $\pi_1 = \pi_1(X, x_0)$. If α, β are paths from x_0 to x_1, then $\hat{\alpha}$ and $\hat{\beta}$ induce the same isomorphism $\pi_1(X, x_0)/[\pi_1, \pi_1] \to \pi_1(X, x_1)/[\pi_1, \pi_1]$: with $g = \alpha * \bar{\beta}$, $\hat{g}([f]) = [g]^{-1}[f][g]$, which has the same coset as $[f]$ in the abelian quotient. This abelian group, determined up to canonical isomorphism, is the **first homology group** $H_1(X)$. A homeomorphism induces an isomorphism of fundamental groups, hence of H_1.

Theorem 12.5 (75.1)

Let $N \lhd F$ and $q : F \to F/N$, $p : F \to F/[F, F]$ the projections. Then p induces an isomorphism $\phi : q(F)/[q(F), q(F)] \to p(F)/p(N)$.

Proof. Let $r : p(F) \to p(F)/p(N)$ and $s : q(F) \to q(F)/[q(F), q(F)]$ be projections. $r \circ p$ kills N, so it induces $u : q(F) \to p(F)/p(N)$, and u kills commutators (the target is abelian), inducing ϕ. Likewise $s \circ q$ maps into an abelian group, so induces $v : p(F) \to q(F)/[q(F), q(F)]$, and $v \circ p = s \circ q$ kills N, so v kills $p(N)$ and induces ψ. Explicitly, $\phi(sq(x)) = rp(x)$ and $\psi(rp(x)) = sq(x)$; they are inverse to each other. ■

Corollary 12.6 (75.2)

Let F be free on $\alpha_1, \dots, \alpha_n$, N the least normal subgroup containing x, and $G = F/N$. Then $G/[G, G]$ is the quotient of the free abelian group $F/[F, F]$ (basis $p(\alpha_i)$) by the subgroup generated by $p(x)$.

Proof. N is generated by the conjugates of x (Lemma 11.17), and all of them have image $p(x)$ in the abelian group, so $p(N)$ is generated by $p(x)$. Apply Theorem 12.5 and Theorem 11.21. ■

Theorem 12.7 (75.3)

$H_1(T_n)$ is free abelian of rank $2n$.

Proof. By Theorem 12.3 and Corollary 12.6, $H_1(T_n)$ is the free abelian group on α_i, β_i modulo the image of $[\alpha_1, \beta_1] \cdots [\alpha_n, \beta_n]$, which is 0 in an abelian group. ■

Theorem 12.8 (75.4)

The torsion subgroup $T(P_m)$ of $H_1(P_m)$ has order 2, and $H_1(P_m)/T(P_m)$ is free abelian of rank $m-1$.

Proof. $H_1(P_m)$ is the free abelian group on $\alpha_1, \dots, \alpha_m$ modulo the subgroup generated by $2(\alpha_1 + \cdots + \alpha_m)$. With $\gamma = \alpha_1 + \cdots + \alpha_m$, the elements $\alpha_1, \dots, \alpha_{m-1}, \gamma$ form another basis, so $H_1(P_m) \cong \mathbb{Z}^{m-1} \times \mathbb{Z}/2$. ■

Theorem 12.9 (75.5)

The surfaces $S^2, T_1, T_2, \dots, P_1, P_2, \dots$ are topologically distinct.

Proof. Their first homology groups are 0, $\mathbb{Z}^{2n}$ and $\mathbb{Z}^{m-1} \times \mathbb{Z}/2$, pairwise non-isomorphic (by rank, Theorem 11.8, and by torsion). ■

12.3 Cutting and pasting

Motivation. To show that every compact surface is one of the standard ones, we manipulate polygon gluing schemes by cutting and regluing, step by step, into a normal form.

Let P have vertices $p_0, \dots, p_{n-1}$ and $1 < k < n-1$. Cutting P along p_0p_k gives Q_1 (vertices $p_0, \dots, p_k$) and Q_2 (vertices $p_k, \dots, p_n = p_0$); after translating Q_1 to be disjoint from Q_2, P is homeomorphic to the quotient of $Q_1 \sqcup Q_2$ pasting the two copies of p_0p_k linearly. Conversely, two disjoint regions pasted along one edge each can be represented by one region: move the vertices of one onto the circle of the other.

Theorem 12.10 (76.1)

Let X be obtained from m regions by the scheme $y_0y_1, w_2, \dots, w_m$, and let c be a label not appearing in it. If y_0 and y_1 have length at least 2, then X is also obtained from $m+1$ regions by the scheme $y_0c^{-1}, cy_1, w_2, \dots, w_m$. Conversely, the second scheme gives the same space as the first.

Proof. Cut the first region along the diagonal from p_0 to p_k, k the length of y_0, and label both copies of the diagonal c: it runs counterclockwise in the second piece and clockwise in the first. A composite of quotient maps is a quotient map, so pasting the two copies of c first and then the other edges gives the same space as pasting everything at once. ■

The following **elementary operations** on a scheme do not change the space. Two schemes are **equivalent** if one is obtained from the other by such operations; each operation has an inverse operation.

(i) *Cut*: replace y_0y_1 by y_0c^{-1} and cy_1 (c new, y_0, y_1 of length ≥ 2).

(ii) *Paste*: replace y_0c^{-1} and cy_1 by y_0y_1 (c appearing nowhere else).

(iii) *Relabel*: replace all occurrences of a label by a new label, or change the sign of all exponents of one label.

(iv) *Permute*: replace y_0y_1 by y_1y_0 (renumber the vertices).

(v) *Flip*: replace $(a_{i_1})^{\epsilon_1} \cdots (a_{i_n})^{\epsilon_n}$ by $(a_{i_n})^{-\epsilon_n} \cdots (a_{i_1})^{-\epsilon_1}$ (reflect the region).

(vi) *Cancel*: replace $y_0aa^{-1}y_1$ by y_0y_1 if a appears nowhere else and y_0, y_1 have length ≥ 2. Indeed, cut to get y_0ab and $b^{-1}a^{-1}y_1$; in each region the two edges a, b can be merged into one edge c (the composite pasting is linear on each), giving y_0c and $c^{-1}y_1$; paste.

(vii) *Uncancel*: the reverse of (vi).

Example 12.1

The Klein bottle $aba^{-1}b$ is P_2: cut to abc^{-1} and $ca^{-1}b$; permute the first to $bc^{-1}a$; flip the second to $b^{-1}ac^{-1}$ and permute it to $ac^{-1}b^{-1}$; paste along b to get $ac^{-1}c^{-1}a$; permute and relabel to $aabb$.

12.4 The classification theorem

Motivation. Every compact connected surface is homeomorphic to a sphere, a connected sum of tori, or a connected sum of projective planes, and these are mutually distinct. Topologically, surfaces are classified by orientability and one integer.

A scheme is **proper** if each label appears exactly twice; elementary operations preserve properness. A proper scheme w for one region is of **torus type** if each label appears once with each exponent, and of **projective type** otherwise. In what follows, brackets only group letters, and y^{-1} is the formal inverse of a word y.

Lemma 12.11 (77.1)

If $w = [y_0]a[y_1]a[y_2]$ is proper (some y_i possibly empty), then $w \sim aa[y_0y_1^{-1}y_2]$.

Proof.

Step 1. y_0 empty, $w = ay_1ay_2$. If y_1 is empty there is nothing to do. If y_2 is empty, flip to $a^{-1}y_1^{-1}a^{-1}$, permute and relabel to aay_1^{-1}. Otherwise cut into ay_1c^{-1} and cay_2; flip the first to $cy_1^{-1}a^{-1}$ and permute the second to ay_2c; paste along a to get $cy_1^{-1}y_2c$; permute and relabel to $aay_1^{-1}y_2$.

Step 2. y_0 nonempty. If y_1, y_2 are empty, permute. Otherwise cut w into y_0ac^{-1} and cy_1ay_2; flip the first and permute to $y_0^{-1}ca^{-1}$; permute the second to ay_2cy_1; paste along a: $y_0^{-1}cy_2cy_1$. Permute to $cy_2cy_1y_0^{-1}$; by Step 1 this is equivalent to $ccy_2^{-1}y_1y_0^{-1}$; flip to $y_0y_1^{-1}y_2c^{-1}c^{-1}$; permute and relabel to $aay_0y_1^{-1}y_2$. ∎

Corollary 12.12 (77.2)

A scheme of projective type is equivalent to one of the same length of the form $(a_1a_1)(a_2a_2)\cdots(a_ka_k)w_1$ with $k \geq 1$ and w_1 empty or of torus type.

Proof. Some label occurs twice with the same exponent, which we may take to be $+1$; Lemma 12.11 gives $w \sim aaw_1$. If w_1 is not of torus type, write $w = aaz_0bz_1bz_2$ and apply the lemma with $y_0 = aaz_0$: $w \sim bbaaz_0z_1^{-1}z_2$. Each step lengthens the initial block of pairs, so the process ends with w_1 empty or of torus type. ∎

In the next proof we use one more consequence of the operations.

Step E. *If a does not occur in B, D_1, D_2, then the cyclic scheme $aBa^{-1}D_1D_2$ is equivalent to $aBa^{-1}D_2D_1$.* We may assume D_1, D_2 nonempty. Permute to $Ba^{-1}D_1D_2a$, cut into $Ba^{-1}D_1c^{-1}$ and cD_2a, permute these to $D_1c^{-1}Ba^{-1}$ and acD_2, paste along a: $D_1c^{-1}BcD_2$. Permuting and relabelling $(c \to a^{-1})$ gives $aBa^{-1}D_2D_1$. Writing $aBa^{-1}D$ cyclically as $a^{-1}DaB$ and relabelling a, the same applies to the block B: *either block between a and a^{-1} may be cyclically rotated.*

Lemma 12.13 (77.3)

Let $w = w_0 w_1$ be proper with w_1 of torus type containing no two adjacent terms with the same label. Then $w \sim w_0 w_2$ with w_2 of the same length as w_1 and of the form $aba^{-1}b^{-1}w_3$, w_3 empty or of torus type.

Proof. Among the labels of w_1 let a be one whose two occurrences are closest; they are not adjacent, and we may assume a comes first. Let b occur between them, with exponent $+1$ after relabelling. Its other occurrence b^{-1} is not between a and a^{-1}. If it follows a^{-1}, then $w_1 = y_1 aPbQa^{-1}Rb^{-1}y_5$; if it precedes a, exchange the roles of a and b (after changing exponents) to reach the same form. Cyclically, $w = aPbQa^{-1}Rb^{-1}S$ with $S = y_5 w_0 y_1$. Using Step E repeatedly: rotate the block PbQ between a and a^{-1} to bQP, obtaining $ab(QP)a^{-1}Rb^{-1}S$; rotate the block $(QP)a^{-1}R$ between b and b^{-1} to $a^{-1}RQP$, obtaining $aba^{-1}(RQP)b^{-1}S$; rotate the block $(RQP)b^{-1}S$ following a^{-1} to $b^{-1}SRQP$, obtaining $aba^{-1}b^{-1}T$ with $T = SRQP$. Finally, for any splitting $T = T_1T_2$, $aba^{-1}b^{-1}T_1T_2 \sim aba^{-1}b^{-1}T_2T_1$: rotate the block T_1T_2a between b^{-1} and b to T_2aT_1 and repeat the three rotations. With $T_1 = y_5 w_0$ and $T_2 = y_1 RQP$ we get, after permuting, $w_0 aba^{-1}b^{-1}w_3$ where $w_3 = y_1 RQP y_5$ consists of the remaining letters of w_1 with unchanged exponents, so it is empty or of torus type. ∎

Lemma 12.14 (77.4)

If $w = w_0(cc)(aba^{-1}b^{-1})w_1$ is proper, then $w \sim w_0 aabbccw_1$.

Proof. Permute to $cc[ab][ba]^{-1}[w_1w_0]$. By Lemma 12.11 read backwards (with $y_0 = ab$, $y_1 = ba$, $y_2 = w_1w_0$), this is equivalent to $[ab]c[ba]c[w_1w_0] = [a]b[c]b[acw_1w_0]$, which by the lemma is equivalent to $bb[ac^{-1}acw_1w_0] = [bb]a[c^{-1}]a[cw_1w_0]$, and again to $aa[bbccw_1w_0]$. Permute. ∎

Theorem 12.15 (77.5, classification theorem)

A space obtained from a polygonal region by pasting its edges together in pairs is homeomorphic to S^2, to T_n, or to P_m.

Proof. The scheme w is proper of length at least 4. We show w is equivalent to one of (1) $aa^{-1}bb^{-1}$ (S^2), (2) $abab$ (P^2), (3) $(a_1a_1)\cdots(a_ma_m)$, $m \geq 2$ (P_m), (4) $(a_1b_1a_1^{-1}b_1^{-1})\cdots(a_nb_na_n^{-1}b_n^{-1})$ (T_n). If w has length greater than 4 and contains adjacent terms aa^{-1}, cancelling (after permuting so that both remaining parts have length ≥ 2) gives a shorter scheme of the same type.

Step 1 (torus type). A torus-type scheme of length 4 is, up to permuting and relabelling, $aa^{-1}bb^{-1}$ or $aba^{-1}b^{-1}$. For longer w we use induction; we may assume no cancellation is possible. Lemma 12.13 gives $w \sim aba^{-1}b^{-1}w_3$ with w_3 nonempty of torus type, and again with no adjacent pair; applying the lemma with $w_0 = aba^{-1}b^{-1}$, and so on, we reach type (4).

Step 2 (projective type). For length 4, Corollary 12.12 gives $aabb$ (type 3) or aay with $y = b^{\pm 1}b^{\mp 1}$; writing $y = y_1^{-1}y_2$ with single letters, Lemma 12.11 gives ay_1ay_2, which relabels to $abab$. For longer w, Corollary 12.12 gives $(a_1a_1)\cdots(a_ka_k)w_1$. If w_1 is empty we have type (3). If w_1 has an adjacent pair, cancel and use induction. Otherwise Lemma 12.13 gives $(a_1a_1)\cdots(a_ka_k)aba^{-1}b^{-1}w_2$, and Lemma 12.14 turns this into $(a_1a_1)\cdots(a_{k-1}a_{k-1})aabba_ka_kw_2$, with a shorter torus part. Repeat. ∎

12.5 Constructing compact surfaces

Motivation. The classification assumed that every compact surface comes from a polygon; this section explains why, using triangulations.

Let X be compact Hausdorff. A **curved triangle** in X is a subspace A with a homeomorphism $h : T \to A$ from a closed triangular region; edges and vertices of A are images of those of T. A **triangulation** of X is a finite collection of curved triangles covering X such that two distinct ones meet in nothing, a common vertex, or a common edge, and in the last case $h_j^{-1} \circ h_i$ is linear on the corresponding edge of T_i. Every compact surface is triangulable (Radó); we assume this.

Theorem 12.16 (78.1)

A compact triangulable surface is homeomorphic to the space obtained from finitely many disjoint triangular regions by pasting their edges together in pairs.

Proof. Let $A_1, \dots, A_n$ be a triangulation with $h_i : T_i \to A_i$, T_i disjoint. Then $h : E = \bigsqcup T_i \to X$ is a closed continuous surjection, hence a quotient map. It suffices to show (a) each edge of each A_i is an edge of exactly one other triangle, and (b) if $A_i \cap A_j$ is a vertex v, there is a chain of triangles with vertex v from A_i to A_j, consecutive ones sharing an edge. Then h identifies exactly the points identified by the edge pastings.

Step 1 (b). Call two triangles with vertex v equivalent if joined by such a chain. If there were two classes, let B be the union of one and C of the rest; $B \cap C = \{v\}$. For any neighbourhood W of v missing the triangles not containing v, $W - v$ is the union of the disjoint relatively closed nonempty sets $(W \cap B) - v$ and $(W \cap C) - v$, so it is not connected. But v has a neighbourhood homeomorphic to an open disc and hence arbitrarily small neighbourhoods W with $W - v$ connected.

Step 2 (at least one other triangle). Let x be interior to an edge of a triangular region T. Its ϵ-neighbourhoods W in T have $W - x$ star-shaped about a point interior to T, hence simply connected. On the other hand, if x had a neighbourhood U homeomorphic to the open unit disc B with $x \mapsto 0$, then for every neighbourhood $V \subset U$ of x, choosing $B_\epsilon \subset V$, the inclusion $B_\epsilon - 0 \to B - 0$ is homotopic to the homeomorphism $y \mapsto y/\epsilon$, so it induces an isomorphism and $\pi_1(V - 0) \to \pi_1(B - 0) \cong \mathbb{Z}$ is surjective: $V - 0$ is not simply connected. Contradiction. Now if an edge e of A_i lay in no other triangle, a point x interior to e would have a neighbourhood in X contained in A_i, and a disc neighbourhood of x in X would give one in T_i.

Step 3 (at most one). Let $Y \subset \mathbb{R}^3$ be the union of $k \geq 3$ triangles meeting pairwise in a common edge e, and $x = 0$ interior to e. The union A of the edges other than e consists of k arcs with common end points; it retracts onto the union of three of them (map each extra arc homeomorphically onto one of the three, fixing end points), a theta space with nonabelian fundamental group (Example 11.1), so $\pi_1(A)$ is nonabelian. A is a deformation retract of $Y - 0$ (radial projection from 0 in each triangle). For any neighbourhood W of 0 in Y, choose ϵ with $Y' = \epsilon Y \subset W$; the inclusion $Y' - 0 \to Y - 0$ is homotopic to $y \mapsto y/\epsilon$, a homeomorphism, so $\pi_1(W - 0) \to \pi_1(Y - 0)$ is surjective and $\pi_1(W - 0)$ is nonabelian (at a base point in Y'). A disc neighbourhood W of 0 would have $\pi_1(W - 0) \cong \mathbb{Z}$. If e were an edge of three or more triangles of the triangulation, a point interior to e would have a neighbourhood in X contained in their union, which is homeomorphic to such a Y. ∎

Theorem 12.17 (78.2)

A compact connected triangulable surface is homeomorphic to a space obtained from one polygonal region by pasting its edges together in pairs.

Proof. By Theorem 12.16, X comes from triangular regions by a proper scheme. Whenever two different regions have edges with the same label, flip one if necessary and paste them along those edges (operation (ii)), reducing the number of regions. When this is no longer possible, if more than one region remained, each would give a nonempty open and closed subset of X, contradicting connectedness. ∎

Combining Theorems 12.17, 12.15 and 12.9: every compact connected surface is homeomorphic to exactly one of S^2, T_n, P_m.

Chapter 13. Classification of Covering Spaces

Throughout this chapter a covering map $p : E \to B$ is assumed to have E and B path connected and locally path connected. For $p(e_0) = b_0$ let $H_0 = p_*(\pi_1(E, e_0))$, a subgroup of $\pi_1(B, b_0)$ isomorphic to $\pi_1(E, e_0)$ (Theorem 9.17).

13.1 Equivalence of covering spaces

Motivation. Covering spaces were introduced to compute fundamental groups; this chapter goes the other way and classifies covering spaces by subgroups of π_1, in close analogy with Galois theory, where intermediate fields correspond to subgroups.

Covering maps $p : E \to B$ and $p' : E' \to B$ are **equivalent** if there is a homeomorphism $h : E \to E'$ with $p = p' \circ h$ (an **equivalence**).

Lemma 13.1 (79.1, general lifting lemma)

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$, and $f : Y \to B$ continuous with $f(y_0) = b_0$, where Y is path connected and locally path connected. Then f has a lifting $\tilde{f} : Y \to E$ with $\tilde{f}(y_0) = e_0$ iff $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(E, e_0))$. Such a lifting is unique.

Proof. If $\tilde{f}$ exists, $f_* = p_*\tilde{f}_*$.

Uniqueness: for $y_1 \in Y$ and a path α from y_0 to y_1, $\tilde{f} \circ \alpha$ is the lifting of $f \circ \alpha$ from e_0, so $\tilde{f}(y_1)$ is its end point.

Existence: define $\tilde{f}(y_1)$ as the end point of the lifting of $f \circ \alpha$ from e_0. It is well defined: if β is another path from y_0 to y_1, lift $f \circ \alpha$ to γ from e_0 and $f \circ \bar{\beta}$ to δ from $\gamma(1)$; then $\gamma * \delta$ lifts the loop $f \circ (\alpha * \bar{\beta})$, whose class lies in $f_*\pi_1(Y, y_0) \subset H_0$, so it is a loop (Theorem 9.17(c)); then $\bar{\delta}$ is the lifting of $f \circ \beta$ from e_0 and ends at $\gamma(1)$. Continuity at y_1: given a neighbourhood N of $\tilde{f}(y_1)$, choose a path-connected evenly covered $U \ni f(y_1)$ whose slice V_0 through $\tilde{f}(y_1)$ lies in N, and a path-connected neighbourhood W of y_1 with $f(W) \subset U$. For $y \in W$ and a path δ in W from y_1 to y, the lifting of $f \circ (\alpha * \delta)$ is $\gamma * ((p|V_0)^{-1} \circ f \circ \delta)$, ending in V_0. So $\tilde{f}(W) \subset V_0 \subset N$. ∎

Theorem 13.2 (79.2)

Let $p : E \to B$, $p' : E' \to B$ be covering maps with $p(e_0) = p'(e_0') = b_0$. There is an equivalence $h : E \to E'$ with $h(e_0) = e_0'$ iff $H_0 = p_*\pi_1(E, e_0)$ and $H_0' = p_*'\pi_1(E', e_0')$ are equal. Such h is unique.

Proof. If h exists, h_* is an isomorphism and $p = p'h$ gives $H_0 = H_0'$. Conversely, Lemma 13.1 gives liftings $h : E \to E'$ of p with $h(e_0) = e_0'$ and $k : E' \to E$ of p' with $k(e_0') = e_0$. Then $k \circ h$ and id_E both lift p and fix e_0, so they are equal by uniqueness; likewise $h \circ k = \mathrm{id}$. Uniqueness of h is that of liftings. ∎

Lemma 13.3 (79.3)

Let $e_0, e_1 \in p^{-1}(b_0)$ and $H_i = p_*\pi_1(E, e_i)$.
(a) If γ is a path in E from e_0 to e_1 and $\alpha = p \circ \gamma$, then $[\alpha]H_1[\alpha]^{-1} = H_0$.
(b) Conversely, if H is conjugate to H_0, then $H = H_1$ for some $e_1 \in p^{-1}(b_0)$.

Proof. (a) If $[h] \in H_1$, $h = p \circ \tilde{h}$ with $\tilde{h}$ a loop at e_1; $\gamma * \tilde{h} * \bar{\gamma}$ is a loop at e_0 mapping to $[\alpha][h][\alpha]^{-1}$, so $[\alpha]H_1[\alpha]^{-1} \subset H_0$. Using $\bar{\gamma}$, $[\alpha]^{-1}H_0[\alpha] \subset H_1$.

(b) If $H_0 = [\alpha]H[\alpha]^{-1}$, lift α to γ from e_0 and let $e_1 = \gamma(1)$; by (a), $H_0 = [\alpha]H_1[\alpha]^{-1}$, so $H = H_1$. ■

Theorem 13.4 (79.4)

p and p' are equivalent iff H_0 and H_0' are conjugate in $\pi_1(B, b_0)$.

Proof. If h is an equivalence and $e_1' = h(e_0)$, then $H_0 = p_*'\pi_1(E', e_1')$ by Theorem 13.2, which is conjugate to H_0' by Lemma 13.3. Conversely, if they are conjugate, Lemma 13.3(b) gives e_1' with $p_*'\pi_1(E', e_1') = H_0$, and Theorem 13.2 gives an equivalence carrying e_0 to e_1'. ■

Example 13.1

$\pi_1(S^1) \cong \mathbb{Z}$ is abelian with subgroups $n\mathbb{Z}$. So every covering of S^1 is equivalent to $p : \mathbb{R} \to S^1$ (trivial subgroup) or to $z \mapsto z^n$ (subgroup $n\mathbb{Z}$).

13.2 The universal covering space

Motivation. The simply connected covering space sits above all others, just as the splitting field sits above every intermediate field. Its existence, under mild local conditions, makes the correspondence complete.

If E is simply connected, p is a **universal covering**; by Theorem 13.4 it is unique up to equivalence.

Lemma 13.5 (80.1)

Let B be path connected and locally path connected, and $p : E \to B$ a covering map with E not necessarily path connected. If E_0 is a path component of E, then $p|E_0 : E_0 \to B$ is a covering map.

Proof. E is locally path connected, so E_0 is open and $p(E_0)$ is open. If $x \in \overline{p(E_0)}$, take a path-connected evenly covered $U \ni x$; some slice V meets E_0, and being path connected lies in E_0, so $x \in U = p(V) \subset p(E_0)$. Thus $p(E_0) = B$. For any such U, $(p|E_0)^{-1}(U)$ is the union of the slices meeting E_0, each contained in E_0 and mapped homeomorphically onto U. ■

Lemma 13.6 (80.2)

Let $p = r \circ q$ with $p : X \to Z$, $q : X \to Y$, $r : Y \to Z$ continuous (all spaces path connected and locally path connected).
(a) If p and r are covering maps, so is q.
(b) If p and q are covering maps, so is r.

Proof. Let $x_0 \in X$, $y_0 = q(x_0)$.

(a) q is surjective: for a path $\tilde{\alpha}$ in Y from y_0 to y, lift $r \circ \tilde{\alpha}$ through p to $\tilde{\tilde{\alpha}}$ from x_0; $q \circ \tilde{\tilde{\alpha}}$ lifts $r \circ \tilde{\alpha}$ through r from y_0, so equals $\tilde{\alpha}$, and y is in the image. Given y, take a path-connected $U \ni r(y)$ evenly covered by both p and r, and the slice V of $r^{-1}(U)$ containing y. Each slice U_α of $p^{-1}(U)$ is connected, so q maps it into one slice of $r^{-1}(U)$; $q^{-1}(V)$ is the union of those U_α mapped into V, and on each $q = (r|V)^{-1} \circ (p|U_\alpha)$ is a homeomorphism onto V.

(b) Let U be a path-connected neighbourhood of z evenly covered by p, and V_β the path components of $r^{-1}(U)$, open and disjoint. The slices U_α of $p^{-1}(U)$ are the path components of $p^{-1}(U)$, and q

maps each into one V_β; so $q^{-1}(V_\beta)$ is a union of slices and, q being surjective, contains some U_α. By Theorem 9.10, q restricts to a covering map $q^{-1}(V_\beta) \to V_\beta$, and U_α is a path component of $q^{-1}(V_\beta)$, so $q|U_\alpha : U_\alpha \to V_\beta$ is surjective by Lemma 13.5. It is also open and injective (as $p|U_\alpha$ is injective). Thus $q|U_\alpha$ is a homeomorphism onto V_β, and $r|V_\beta = (p|U_\alpha) \circ (q|U_\alpha)^{-1}$ is a homeomorphism onto U. ■

Theorem 13.7 (80.3)

If $p : E \to B$ is a covering map with E simply connected, then for any covering map $r : Y \to B$ there is a covering map $q : E \to Y$ with $r \circ q = p$.

Proof. Lemma 13.1 applies trivially and gives q with $r \circ q = p$; Lemma 13.6(a) shows q is a covering map. ■

Lemma 13.8 (80.4)

If $p : E \to B$ is a covering map with E simply connected and $p(e_0) = b_0$, then b_0 has a neighbourhood U such that $i_* : \pi_1(U, b_0) \to \pi_1(B, b_0)$ is trivial.

Proof. Let U be evenly covered and $\tilde{U}$ the slice containing e_0. A loop f in U lifts to a loop in $\tilde{U}$, which is nulhomotopic in E; applying p gives a path homotopy in B from f to a constant. ■

Example 13.2

The infinite earring X (Example 11.2) is path connected and locally path connected but has no universal covering: for any neighbourhood U of the origin, some circle C_n lies in U, and C_n is a retract of X, so $\pi_1(C_n) \to \pi_1(X)$ is injective and $i_* : \pi_1(U) \to \pi_1(X)$ is nontrivial.

13.3 Covering transformations

Motivation. The symmetries of a covering (deck transformations) form a group that, for the universal cover, is isomorphic to $\pi_1(B)$. This is the topological counterpart of the Galois group.

An equivalence of p with itself is a **covering transformation**; they form the group $\mathcal{C}(E, p, B)$. The **normalizer** of H in G is $N(H) = \{g : gHg^{-1} = H\}$, the largest subgroup in which H is normal. Let $F = p^{-1}(b_0)$, $\Phi : \pi_1(B, b_0)/H_0 \to F$ the bijection induced by the lifting correspondence (Theorem 9.17), and $\Psi : \mathcal{C}(E, p, B) \to F$, $\Psi(h) = h(e_0)$, injective by uniqueness of liftings.

Lemma 13.9 (81.1)

The image of Ψ equals $\Phi(N(H_0)/H_0)$.

Proof. Let γ be the lifting of a loop α from e_0, $e_1 = \gamma(1) = \Phi([\alpha]H_0)$, $H_1 = p_*\pi_1(E, e_1)$. A covering transformation with $h(e_0) = e_1$ exists iff $H_1 = H_0$ (Theorem 13.2). Since $[\alpha]H_1[\alpha]^{-1} = H_0$ (Lemma 13.3), this holds iff $[\alpha]H_0[\alpha]^{-1} = H_0$, i.e. $[\alpha] \in N(H_0)$. ■

Theorem 13.10 (81.2)

$\Phi^{-1} \circ \Psi : \mathcal{C}(E, p, B) \to N(H_0)/H_0$ is a group isomorphism.

Proof. It is a bijection by Lemma 13.9. Let $h(e_0) = e_1$, $k(e_0) = e_2$, and choose paths γ, δ from e_0 to e_1, e_2; with $\alpha = p \circ \gamma$, $\beta = p \circ \delta$, $\Phi([\alpha]H_0) = e_1$ and $\Phi([\beta]H_0) = e_2$. Then $h \circ \delta$ runs from e_1 to $e_3 = h(k(e_0))$ and lifts β, so $\gamma * (h \circ \delta)$ lifts $\alpha * \beta$ from e_0 and ends at e_3: $\Phi([\alpha * \beta]H_0) = \Psi(h \circ k)$. ■

Corollary 13.11 (81.3)

H_0 is normal in $\pi_1(B, b_0)$ iff for all $e_1, e_2 \in p^{-1}(b_0)$ some covering transformation carries e_1 to e_2. Then $\mathcal{C}(E, p, B) \cong \pi_1(B, b_0)/H_0$.

Proof. If H_0 is normal, Ψ is surjective, giving h_i with $h_i(e_0) = e_i$; take $h_2 h_1^{-1}$. Conversely, transitivity makes Ψ surjective, so $N(H_0)/H_0 = \pi_1/H_0$ and $N(H_0) = \pi_1(B, b_0)$. ■

Corollary 13.12 (81.4)

If E is simply connected, $\mathcal{C}(E, p, B) \cong \pi_1(B, b_0)$.

If H_0 is normal, p is **regular**. Then $\mathcal{C}$ is transitive on every fibre $p^{-1}(b)$: for $x_1, x_2 \in p^{-1}(b)$ and a path β from b to b_0, lift β from x_1, x_2 to paths ending at e_1, e_2, and let $h(e_1) = e_2$; then h carries the first lifting to a lifting of β ending at e_2, which by uniqueness (lift $\bar{\beta}$ from e_2) is the second, so $h(x_1) = x_2$.

Example 13.3

Every covering of S^1 is regular; for $\mathbb{R} \to S^1$ the covering transformations are $x \mapsto x + n$. For the covering of the figure eight in Lemma 9.45, $\mathcal{C}$ is trivial: a loop around A lifts to a non-loop from the origin but to a loop from every other point of the fibre on the y-axis, and similarly for B, so every covering transformation fixes the origin.

For a group G of homeomorphisms of X, the **orbit space** X/G is the quotient by $x \sim g(x)$. The action is **properly discontinuous** if each x has a neighbourhood U with $g(U) \cap U = \emptyset$ for $g \neq e$; then $g_0(U) \cap g_1(U) = \emptyset$ for $g_0 \neq g_1$.

Theorem 13.13 (81.5)

Let X be path connected and locally path connected and G a group of homeomorphisms of X. The quotient map $\pi : X \to X/G$ is a covering map iff the action is properly discontinuous. In that case π is regular and G is its group of covering transformations.

Proof. π is open, since $\pi^{-1}\pi(U) = \bigcup_g g(U)$. If the action is properly discontinuous and U is as in the definition, $\pi^{-1}\pi(U)$ is the disjoint union of the open sets $g(U)$, each meeting each orbit at most once; $\pi|g(U)$ is a continuous open bijection onto $\pi(U)$, hence a homeomorphism. Conversely, if π is a covering, let U be the slice through x over an evenly covered neighbourhood of $\pi(x)$; if $g \neq e$ and $u' = g(u) \in U$ with $u \in U$, then $u' = u$ by injectivity of $\pi|U$; so g fixes u, and g is a covering transformation fixing a point, hence the identity (shown next), a contradiction. Each g is a covering transformation. If h is one with $h(x_1) = x_2$, then x_1, x_2 lie in one orbit, $g(x_1) = x_2$ for some g, and $h = g$ by Theorem 13.2. G is transitive on fibres, so π is regular (Corollary 13.11). ■

Theorem 13.14 (81.6)

If $p : X \to B$ is regular with group of covering transformations G, there is a homeomorphism $k : X/G \to B$ with $p = k \circ \pi$.

Proof. p is constant on orbits, so it induces a continuous k. Conversely p is a quotient map (continuous, open, surjective), and π is constant on each fibre of p since G is transitive on fibres; so π induces a continuous inverse $B \to X/G$. ■

Example 13.4

On the cylinder $X = S^1 \times I$, $h(x,t) = (-x,t)$ and $k(x,t) = (-x, 1-t)$ generate properly discontinuous actions of $\mathbb{Z}/2$; $X/\langle h\rangle$ is again a cylinder, $X/\langle k\rangle$ is the Möbius band. $\mathbb{Z}^2$ acting on $\mathbb{R}^2$ by translations gives the torus; $\{\pm 1\}$ acting on S^n gives P^n.

13.4 Existence of covering spaces

Motivation. Every subgroup of $\pi_1(B)$ is realized by a covering space, provided B is semilocally simply connected. This completes the classification.

B is **semilocally simply connected** if each b has a neighbourhood U such that $\pi_1(U,b) \to \pi_1(B,b)$ is trivial. By Lemma 13.8 this is necessary for a universal covering.

Theorem 13.15 (82.1)

Let B be path connected, locally path connected and semilocally simply connected, $b_0 \in B$, and H a subgroup of $\pi_1(B,b_0)$. There is a covering map $p : E \to B$ and $e_0 \in p^{-1}(b_0)$ with $p_*\pi_1(E,e_0) = H$.

Proof.

Step 1. Let $\mathcal{P}$ be the set of paths in B from b_0. Put $\alpha \sim \beta$ if $\alpha(1) = \beta(1)$ and $[\alpha * \bar\beta] \in H$, an equivalence relation; write $\alpha^\#$ for the class, E for the set of classes, and $p(\alpha^\#) = \alpha(1)$. Note:

(1) $[\alpha] = [\beta]$ implies $\alpha^\# = \beta^\#$;

(2) $\alpha^\# = \beta^\#$ implies $(\alpha*\delta)^\# = (\beta*\delta)^\#$ for any path δ from $\alpha(1)$, since $[(\alpha*\delta)*\overline{(\beta*\delta)}] = [\alpha*\bar\beta]$.

Step 2. For $\alpha \in \mathcal{P}$ and a path-connected neighbourhood U of $\alpha(1)$ let $B(U,\alpha) = \{(\alpha*\delta)^\# : \delta \text{ a path in } U \text{ from } \alpha(1)\}$; it contains $\alpha^\#$. If $\beta^\# \in B(U,\alpha)$, then $B(U,\beta) = B(U,\alpha)$: with $\beta^\# = (\alpha*\delta)^\#$, elements $(\beta*\gamma)^\# = (\alpha*(\delta*\gamma))^\#$ lie in $B(U,\alpha)$ by (2), and $\alpha^\# = (\beta*\bar\delta)^\#$ gives the reverse inclusion. Hence if $\beta^\# \in B(U_1,\alpha_1) \cap B(U_2,\alpha_2)$, choosing a path-connected $V \subset U_1 \cap U_2$ containing $\beta(1)$, $B(V,\beta) \subset B(U_1,\beta) \cap B(U_2,\beta) = B(U_1,\alpha_1) \cap B(U_2,\alpha_2)$. So these sets form a basis.

Step 3. p is continuous ($B(U,\alpha) \subset p^{-1}(W)$ when $U \subset W$) and open ($p(B(U,\alpha)) = U$).

Step 4. Let U be a path-connected neighbourhood of b_1 with $\pi_1(U,b_1) \to \pi_1(B,b_1)$ trivial. Then $p^{-1}(U)$ is the union of the sets $B(U,\alpha)$ with $\alpha(1) = b_1$: if $p(\beta^\#) \in U$, choose δ in U from b_1 to $\beta(1)$ and $\alpha = \beta * \bar\delta$; then $\beta^\# = (\alpha*\delta)^\#$. Distinct such sets are disjoint by Step 2. And p is injective on $B(U,\alpha)$: if $(\alpha*\delta_1)(1) = (\alpha*\delta_2)(1)$, then $\delta_1 * \bar\delta_2$ is a loop in U at b_1, nulhomotopic in B, so $[\alpha*\delta_1] = [\alpha*\delta_2]$. Being a continuous open bijection, $p|B(U,\alpha)$ is a homeomorphism onto U.

Step 5. Let e_0 be the class of the constant path. For $\alpha \in \mathcal{P}$ and $c \in I$ let $\alpha_c(t) = \alpha(tc)$, and $\tilde\alpha(c) = \alpha_c^\#$. It lifts α from e_0 to $\alpha^\#$. It is continuous at c: a basic neighbourhood of $\tilde\alpha(c)$ is $B(U,\alpha_c)$, U a path-connected neighbourhood of $\alpha(c)$; for d near c, $\alpha([c,d]) \subset U$, and $[\alpha_d] = [\alpha_c * \delta]$ with δ the reparametrized restriction of α to $[c,d]$ (or $[d,c]$, reversed), so $\tilde\alpha(d) \in B(U,\alpha_c)$.

Step 6. E is path connected by Step 5, and locally path connected since it is locally homeomorphic to B; so p is a covering map.

Step 7. For a loop α at b_0, its lifting from e_0 ends at $\alpha^\#$, which equals e_0 iff $[\alpha] \in H$. By Theorem 9.17(c), $p_*\pi_1(E,e_0) = H$. ■

Corollary 13.16 (82.2)

B has a universal covering space iff it is path connected, locally path connected and semilocally simply connected.

Proof. Necessity: Lemma 13.8 (each point of B is in the image of a point of E).

Sufficiency: Theorem 13.15 with H trivial. ■

Together with Theorem 13.4: for such B, equivalence classes of coverings correspond bijectively to conjugacy classes of subgroups of $\pi_1(B, b_0)$.

Chapter 14. Applications to Group Theory

Covering space theory gives proofs of algebraic facts about free groups by realizing free groups as fundamental groups of graphs. (The source PDF does not contain this chapter, so results are named rather than numbered.)

14.1 Covering spaces of a graph

Motivation. As an application of covering-space theory to algebra, we use graphs, whose fundamental groups are free. Covering spaces of graphs are again graphs.

A **linear graph** is a space X that is the union of arcs A_α (**edges**), any two meeting in at most a common end point (a **vertex**), whose topology is coherent with the edges: $C \subset X$ is closed iff $C \cap A_\alpha$ is closed in A_α for every α. A **subgraph** is a union of edges and vertices; it is closed. Fix for each edge a homeomorphism with $[0,1]$. For a subgraph S, the **open star** $\mathrm{St}(S)$ consists of S and the points of edges not in S at parameter distance less than $\frac{1}{2}$ from an end point lying in S; it is open by coherence. We use the standard fact that for compact Hausdorff I, the topology of $X \times I$ is coherent with the subspaces $A_\alpha \times I$ (a product of a quotient map with the identity of a locally compact Hausdorff space is a quotient map).

Lemma 14.1 (graphs are nice)

A linear graph X is Hausdorff and locally path connected; every compact subset meets the interiors of only finitely many edges; and each subgraph S is a deformation retract of $\mathrm{St}(S)$. In particular X is semilocally simply connected.

Proof. Interiors of edges are open in X, and the open stars $\mathrm{St}_\epsilon(v)$ of a vertex (points within parameter distance ϵ of v on the edges at v) are open by coherence; they are path connected, and these sets give enough small neighbourhoods to separate points and to show local path connectedness. If C is compact, pick one point of C in the interior of each edge where possible; every subset of the resulting set D meets each edge in at most one point, so D is closed and discrete in C, hence finite. Pushing each half-open segment of $\mathrm{St}(S) - S$ linearly to its end point in S is continuous on each $(A_\alpha \cap \mathrm{St}(S)) \times I$, hence on $\mathrm{St}(S) \times I$ by coherence, and is a deformation retraction onto S. For a vertex v, $\mathrm{St}(v)$ is contractible; for an interior point of an edge, a small open arc is; so loops in these neighbourhoods are nulhomotopic. ∎

Theorem 14.2 (coverings of graphs)

If $p : E \to X$ is a covering map and X a linear graph, then E is a linear graph whose edges are the liftings of the edges of X, and p maps each edge of E homeomorphically onto an edge of X.

Proof. Parametrize an edge A of X by an injective path $f : I \to A$. For each $e \in p^{-1}(f(0))$, the lifting $\tilde{f}$ from e is injective (as $p \circ \tilde{f} = f$ is), so its image is an arc mapped homeomorphically onto A. Each point of $p^{-1}(A)$ lies on exactly one such lifted arc, by uniqueness of path liftings. Lifted arcs over different edges meet at most in points over vertices, which are end points of both; two liftings of the same edge are disjoint. It remains to see that E has the coherent topology. Let $D \subset E$ meet each lifted edge in a closed set, and let $e \in E$. Choose an evenly covered neighbourhood U of $p(e)$ that is a small open star of a vertex or a small open arc in an edge, and the slice $V \ni e$. Each segment of U along one edge is connected, so it lifts into V inside a single lifted edge; thus $p(D \cap V)$ meets each such segment in a closed set, and since U has the topology coherent with these

segments, $p(D \cap V)$ is closed in U. Hence $D \cap V$ is closed in V. So each point of $E - D$ has a neighbourhood missing D, and D is closed. ■

14.2 The fundamental group of a graph

Motivation. A connected graph deformation retracts to a wedge of circles after collapsing a maximal tree, so its fundamental group is free, with one generator for each edge outside the tree.

A **tree** is a connected linear graph containing no simple closed curve.

Lemma 14.3 (finite trees)

A finite tree with $n \geq 1$ edges has a vertex lying on only one edge; it deformation retracts onto a vertex; and if it has V vertices then it has $V - 1$ edges.

Proof. Take a maximal sequence of distinct edges forming an arc $e_1 \cup \cdots \cup e_k$ (possible since the tree is finite). Its last end point w lies on no other edge: such an edge would either extend the arc or, meeting the arc again, create a simple closed curve. Removing e_k (except its other end point) leaves a tree T' with $n - 1$ edges and one fewer vertex, and collapsing e_k linearly onto that end point is a deformation retraction of T onto T'. Induct. ■

Theorem 14.4 (trees are simply connected)

Every tree T is simply connected.

Proof. A loop f in T has compact image, which meets the interiors of finitely many edges (Lemma 14.1); let Y be the union of these edges and their end points, and Y_0 the component of Y containing $f(I)$. Y_0 is a finite connected subgraph without simple closed curves, a finite tree, so f is nulhomotopic in $Y_0 \subset T$ (Lemma 14.3). ■

Lemma 14.5 (maximal trees)

Every connected linear graph X contains a maximal tree, and a maximal tree contains every vertex of X.

Proof. The union of a chain of trees (subgraphs) is connected and contains no simple closed curve, since such a curve is compact, lies in finitely many edges and hence in one member of the chain. Zorn's lemma gives a maximal tree T. Suppose a vertex is not in T. If every edge with an end point in T had both end points in T, the subgraph W formed by T's vertices and these edges would be open and closed in X (each edge lies in W or misses it), so $W = X$, a contradiction. So some edge A has exactly one end point in T; $T \cup A$ is a tree (a simple closed curve through the new end point would need two edges there), contradicting maximality. ■

Theorem 14.6 (fundamental group of a graph)

Let X be a connected linear graph, T a maximal tree, x_0 a vertex, and $\{A_\alpha\}$ the edges not in T. For each α let f_α be a loop at x_0 that runs in T to one end of A_α, across A_α, and back in T. Then $\pi_1(X, x_0)$ is free with free generators $\{[f_\alpha]\}$.

Proof.

Step 1. finitely many extra edges, $X = T \cup A_1 \cup \cdots \cup A_n$, by induction on n; $n = 0$ is Theorem 14.4. Let $Y = T \cup A_1 \cup \cdots \cup A_{n-1}$, a an interior point of A_n, p, q the end points of A_n, γ the arc in

T from p to q (a finite union of edges), and $C = \gamma \cup A_n$, a simple closed curve. Let $U = X - a$ and $V = \mathrm{St}(C)$ (open stars taken in X). U deformation retracts onto Y (push the two halves of $A_n - a$ to p and q; continuity by the pasting lemma on the closed sets $Y \times I$ and $(A_n - a) \times I$ of $U \times I$). V deformation retracts onto C, and $U \cap V = V - a$ onto the arc $C - a$ (Lemma 14.1). So $U, V, U \cap V$ are path connected, $U \cap V$ is simply connected, and with base point p, Corollary 11.25 gives $\pi_1(X,p) \cong \pi_1(Y,p) * \pi_1(C,p)$. By induction $\pi_1(Y,p)$ is free on the classes of the loops through $A_1, \dots, A_{n-1}$, and $\pi_1(C,p)$ is infinite cyclic generated by the loop around C; by Theorem 11.19 we get the claim at p, and moving the base point along a path in T transforms these loops into the f_α.

Step 2. general case. Any loop or path homotopy in X has compact image, which meets the interiors of finitely many edges, so it lies in $X_F = T \cup \bigcup_{\alpha \in F} A_\alpha$ for some finite F. Hence the $[f_\alpha]$ generate $\pi_1(X, x_0)$, each has infinite order, and no nonempty reduced word in them represents the identity, since otherwise it would do so already in some X_F, contradicting Step 1. ■

Corollary 14.7 (Euler characteristic)

If X is a finite connected linear graph with V vertices and E edges, $\pi_1(X)$ is free of rank $E - V + 1 = 1 - \chi(X)$, where $\chi(X) = V - E$.

Proof. A maximal tree contains all V vertices (Lemma 14.5), hence has $V - 1$ edges (Lemma 14.3), leaving $E - V + 1$ edges outside it. ■

Consequently χ of a finite connected graph depends only on π_1 (by Corollary 11.22), and is unchanged by subdivision.

14.3 Subgroups of free groups

Motivation. Every subgroup of a free group is free (Nielsen–Schreier). The algebraic proof is intricate, but topologically it is immediate: a subgroup of π_1 of a graph is the fundamental group of a covering space, which is again a graph.

A wedge of circles is a linear graph after subdividing each circle into three arcs (the topology, coherent with the circles, is coherent with the arcs). By Lemma 11.30 and Theorem 11.29, every free group is the fundamental group of such a graph.

Theorem 14.8 (Nielsen–Schreier)

Every subgroup H of a free group F is free.

Proof. Let X be a wedge of circles with $\pi_1(X, x_0) \cong F$. X is path connected, locally path connected and semilocally simply connected (Lemma 14.1), so Theorem 13.15 gives a covering $p : E \to X$ with $p_*\pi_1(E, e_0) = H$. As p_* is injective, $H \cong \pi_1(E, e_0)$. E is a connected linear graph (Theorem 14.2), so $\pi_1(E, e_0)$ is free (Theorem 14.6). ■

Theorem 14.9 (Schreier index formula)

If F is free of rank n and H has finite index k in F, then H is free of rank $k(n-1)+1$.

Proof. In the proof above, the fibre $p^{-1}(x_0)$ is in bijection with the right cosets of H (Theorem 9.17), so p is k-sheeted: each vertex and each edge of X has exactly k preimages. The subdivided wedge X has $2n+1$ vertices and $3n$ edges, so $\chi(X) = 1 - n$ and $\chi(E) = k(1-n)$. By Corollary 14.7, $\pi_1(E)$ has rank $1 - \chi(E) = k(n-1)+1$. ■

Example 14.1

(a) Let F be free on a, b, the fundamental group of the figure eight X. The grid of horizontal and vertical lines through integer points of $\mathbb{R}^2$ covers X (Example 9.3); its covering transformations are the integer translations, acting transitively on fibres, so the covering is regular with $F/H \cong \mathbb{Z}^2$, $H = p_*\pi_1(E)$. Since F/H is abelian, $H \supset [F, F]$; the induced surjection $\mathbb{Z}^2 \cong F/[F, F] \to F/H \cong \mathbb{Z}^2$ is injective (a surjective endomorphism of $\mathbb{Z}^2$ has determinant ± 1), so $H = [F, F]$. The grid has the comb (the x-axis and all vertical lines) as a maximal tree, leaving infinitely many edges, so $[F, F]$ is free of infinite rank.
(b) The kernel of $F \to \mathbb{Z}/2$, $a, b \mapsto 1$, has index 2, so it is free of rank $2(2-1)+1 = 3$. Thus, unlike for free abelian groups, a subgroup of a free group can have larger rank than the group.